Solutions: Day 2

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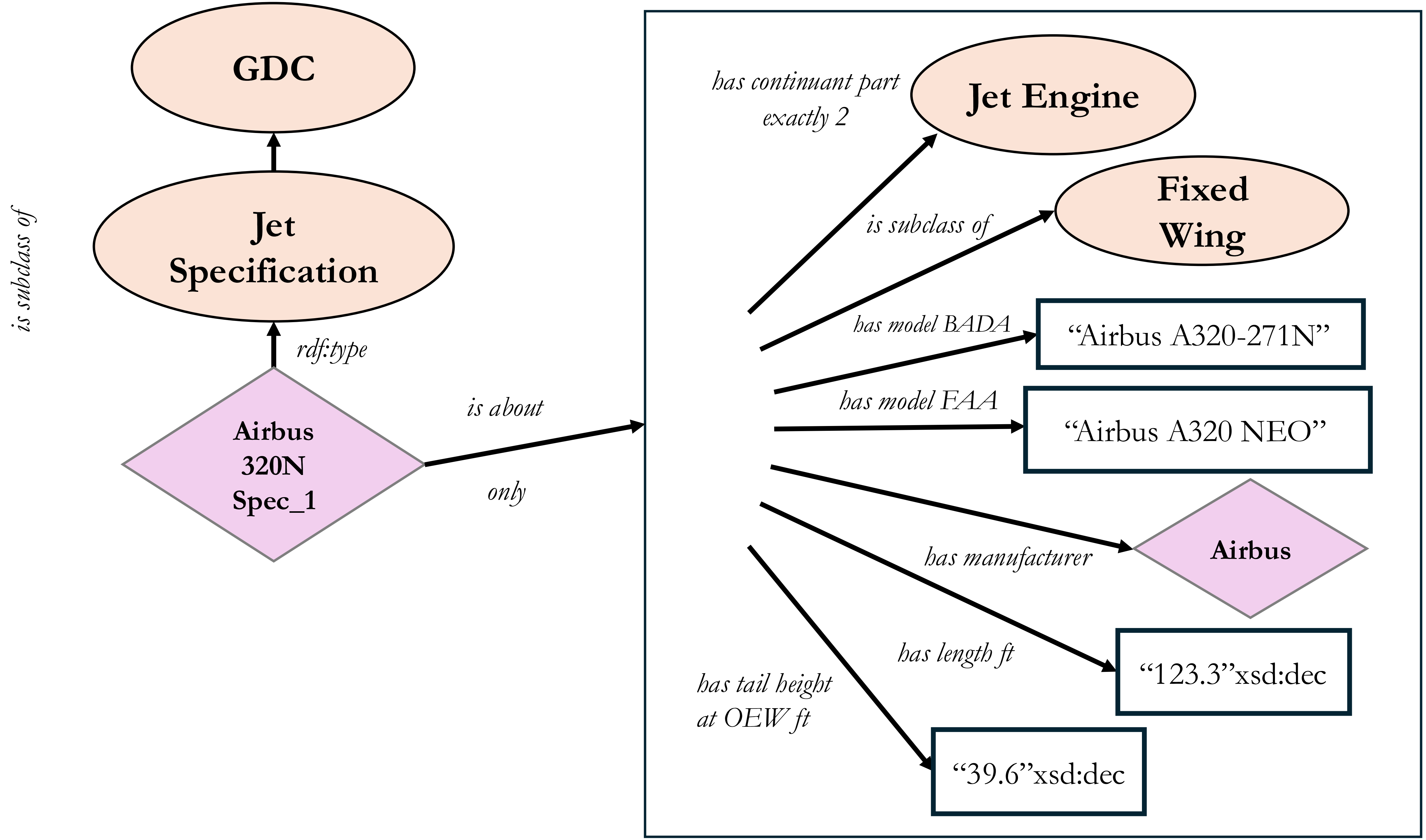
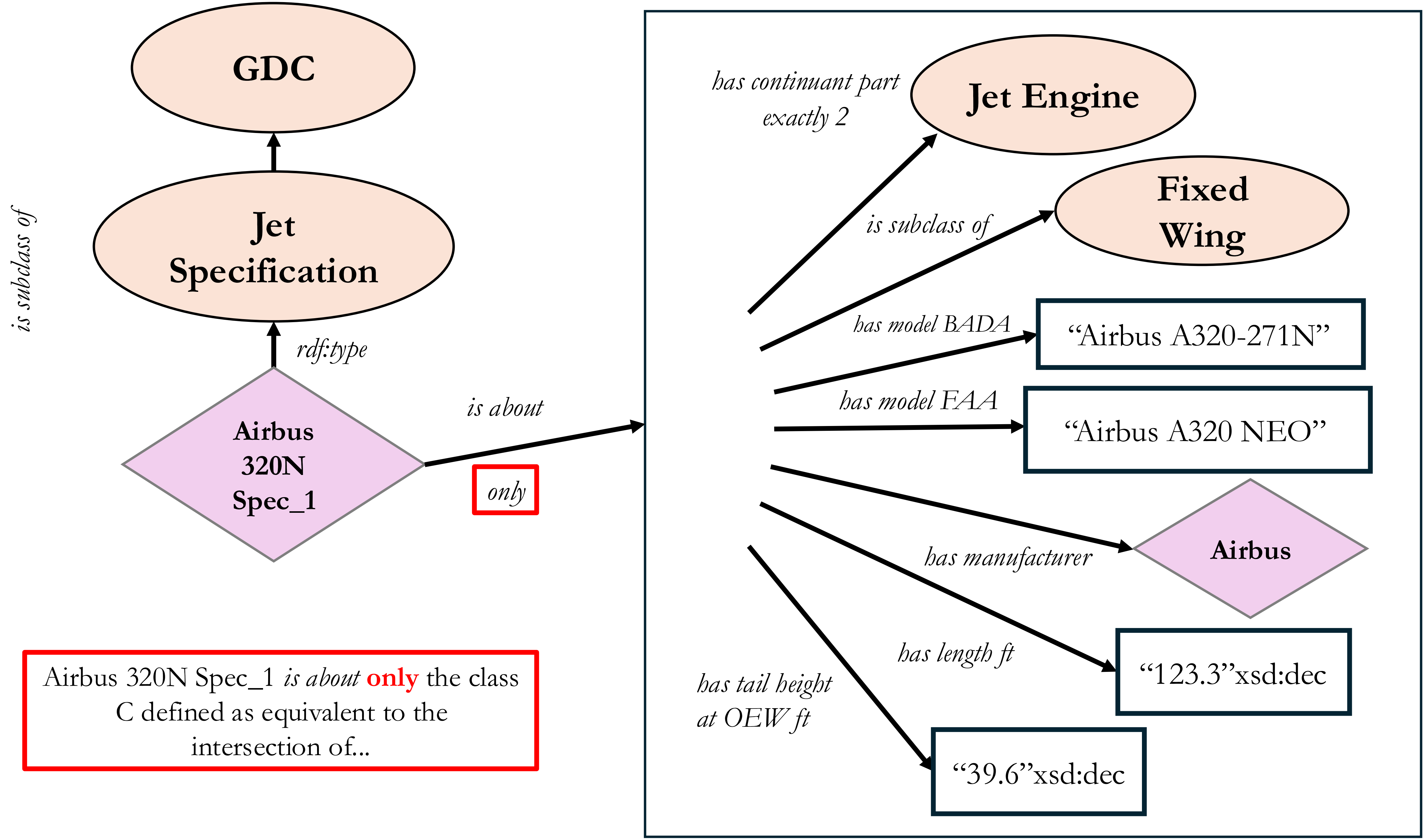
Associate Director, *Institute of AI and Data Science*
CEO, *Acacia Knowledge Systems Inc.*

Case 1

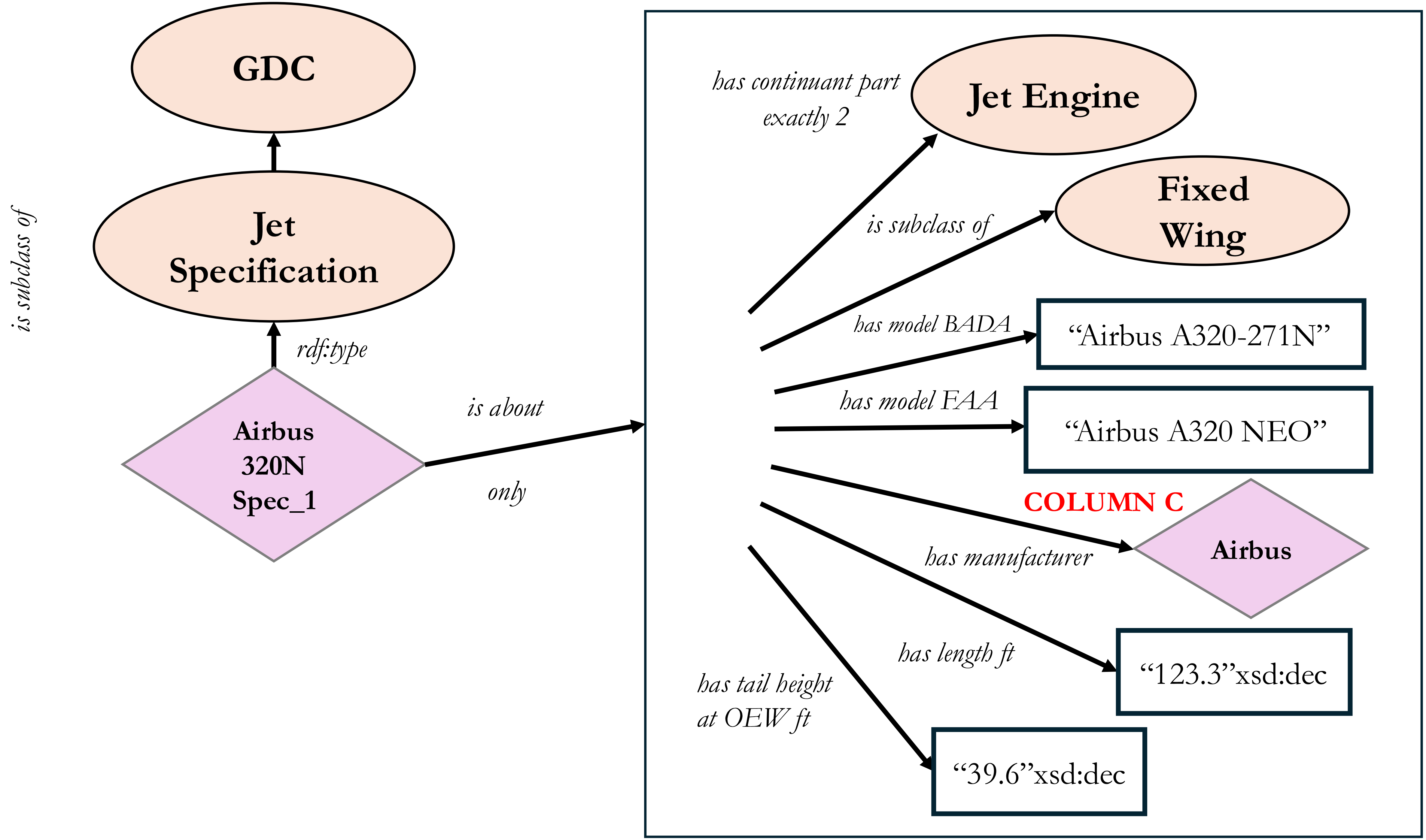
In aircraft_data.xlsx you will find a row for the Airbus A320 Neo.

Construct a design pattern and OWL encoding reflecting the content of columns C-G, R-S, and AB.

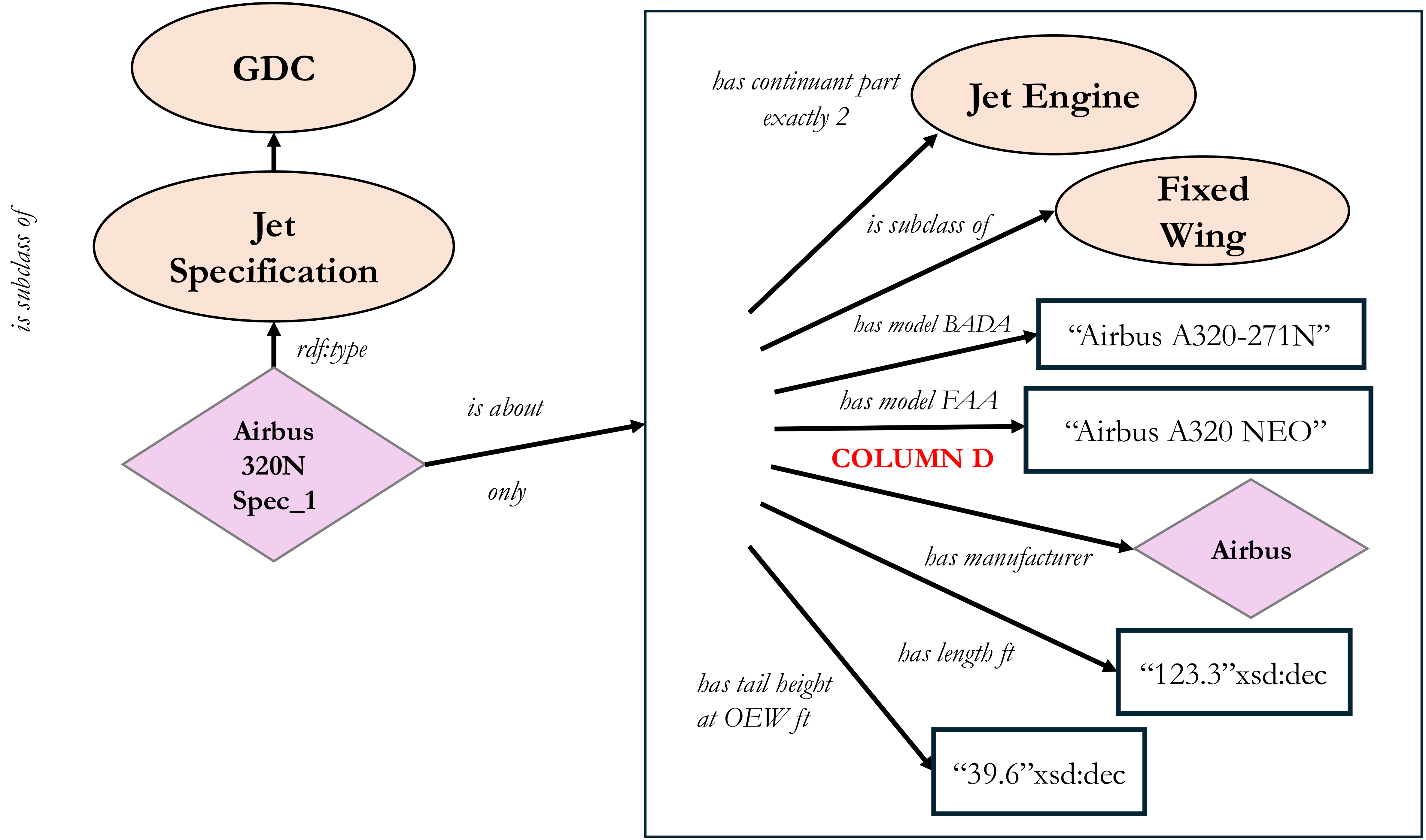

GDC
Jet Specification
is subclass of
rdf:type
Airbus 320N Spec_1
is about
only
has continuant part exactly 2
Jet Engine
is subclass of
Fixed Wing
has model BADA
"Airbus A320-271N"
has model FAA
"Airbus A320 NEO"
has manufacturer
Airbus
has length ft
"123.3"xsd:dec
has tail height at OEW ft
"39.6"xsd:dec



GDC
Jet Specification
is subclass of
rdf:type
Airbus 320N Spec_1
is about
only
Airbus 320N Spec_1 *is about* **only** the class C defined as equivalent to the intersection of...
has continuant part exactly 2
Jet Engine
is subclass of
Fixed Wing
has model BADA
"Airbus A320-271N"
has model FAA
"Airbus A320 NEO"
has manufacturer
Airbus
has length ft
"123.3"xsd:dec
has tail height at OEW ft
"39.6"xsd:dec



GDC
Jet Specification
is subclass of
rdf:type
Airbus 320N Spec_1
is about
only
has continuant part exactly 2
Jet Engine
is subclass of
Fixed Wing
has model BADA
"Airbus A320-271N"
has model FAA
"Airbus A320 NEO"
COLUMN C
Airbus
has manufacturer
has length ft
"123.3"xsd:dec
has tail height at OEW ft
"39.6"xsd:dec



GDC
Jet Specification
is subclass of
rdf:type
Airbus 320N Spec_1
is about
only
has continuant part exactly 2
Jet Engine
is subclass of
Fixed Wing
has model BADA
"Airbus A320-271N"
has model FAA
"Airbus A320 NEO"
COLUMN D
has manufacturer
Airbus
has length ft
"123.3"xsd:dec
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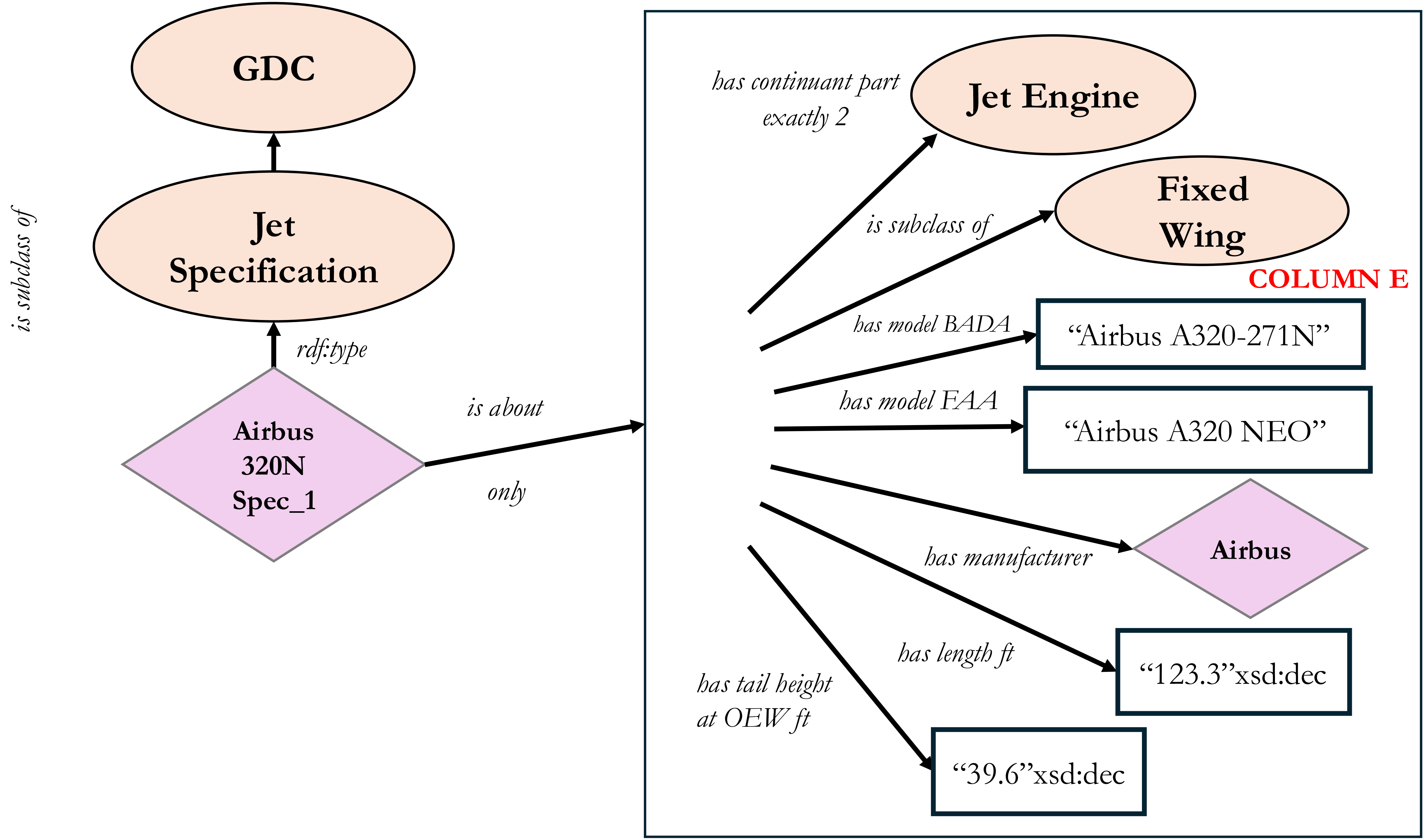
"39.6"xsd:dec


GDC
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is subclass of
rdf:type
Airbus 320N Spec_1
is about
only
has continuant part exactly 2
Jet Engine
is subclass of
Fixed Wing
COLUMN E
has model BADA
"Airbus A320-271N"
has model FAA
"Airbus A320 NEO"
has manufacturer
Airbus
has length ft
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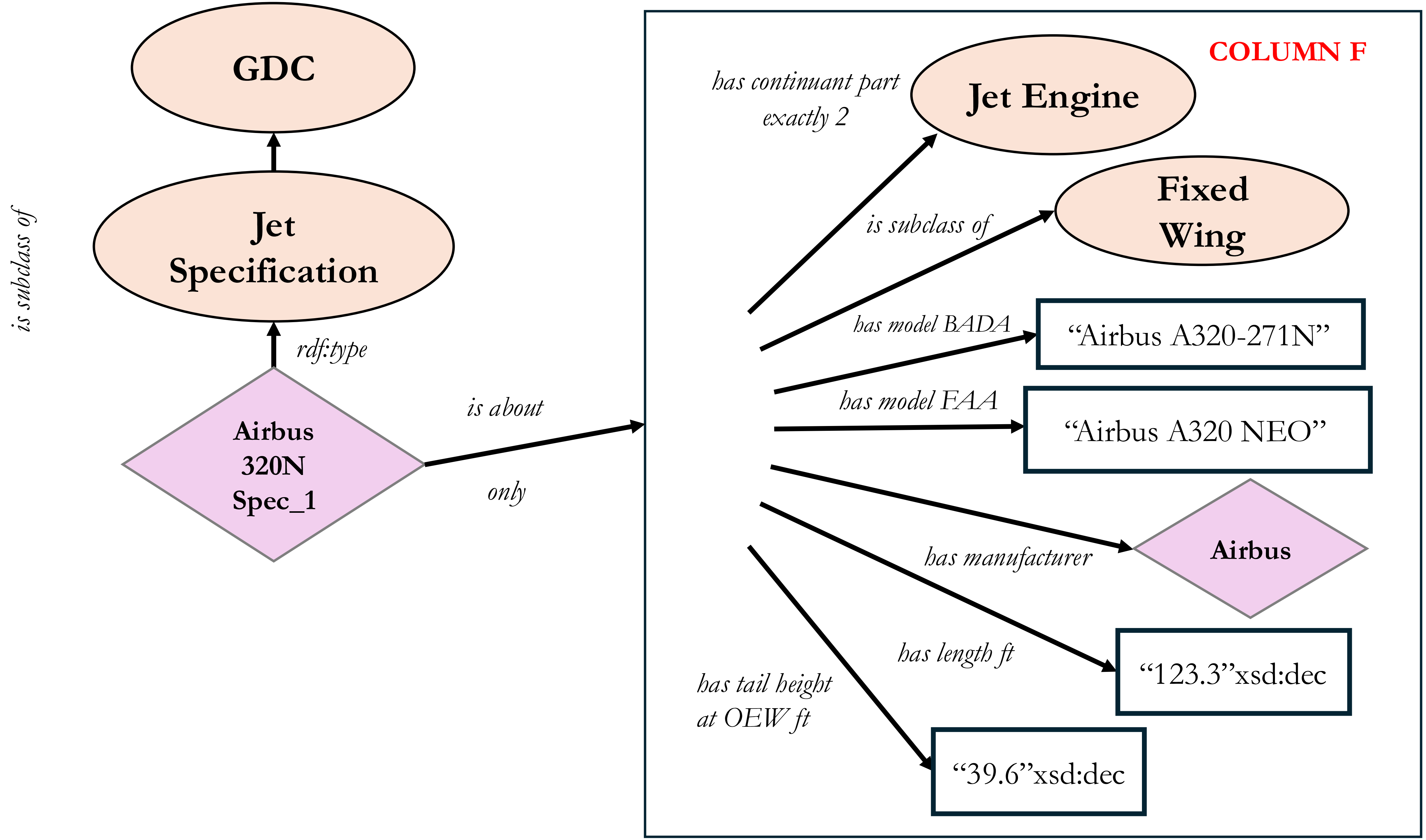
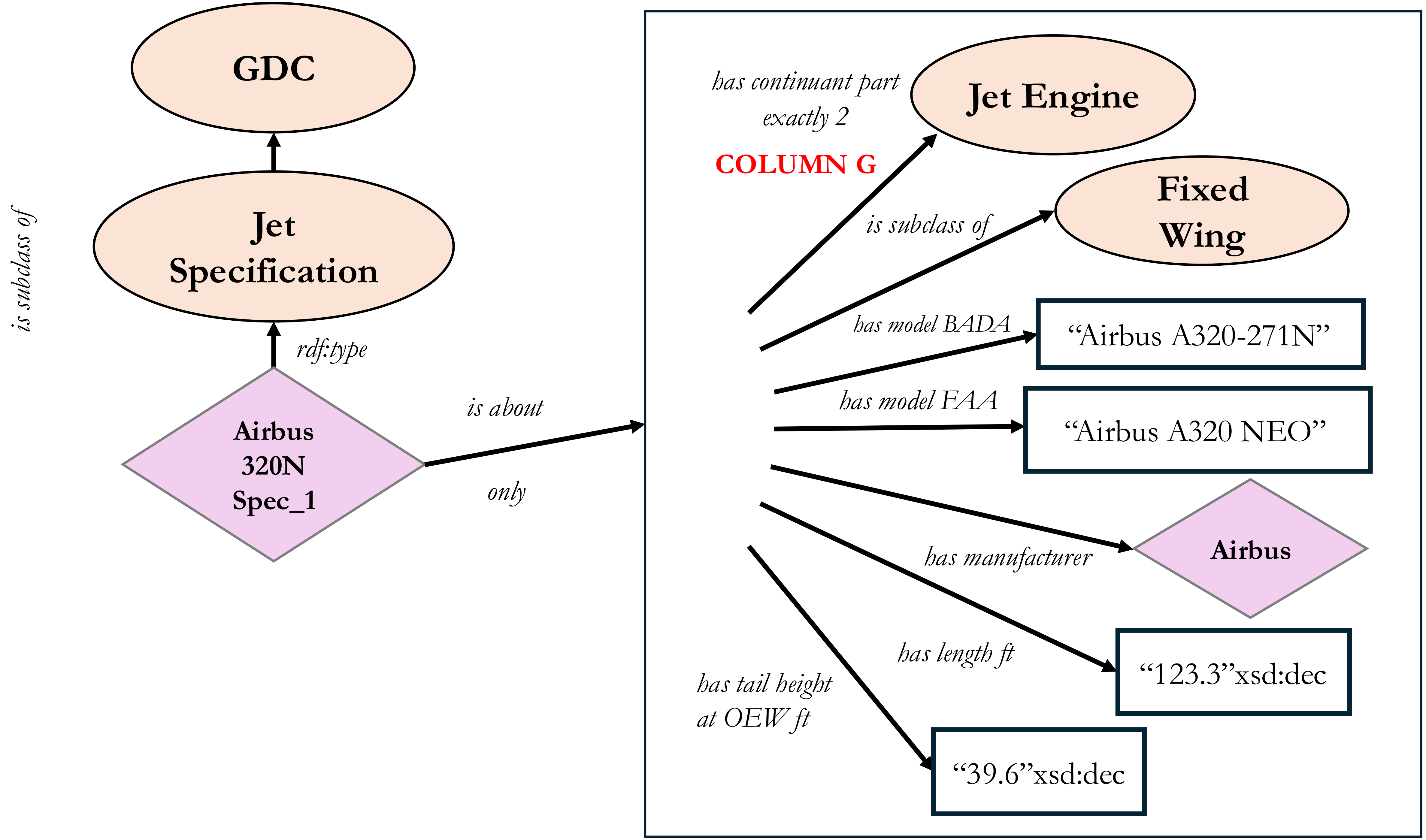
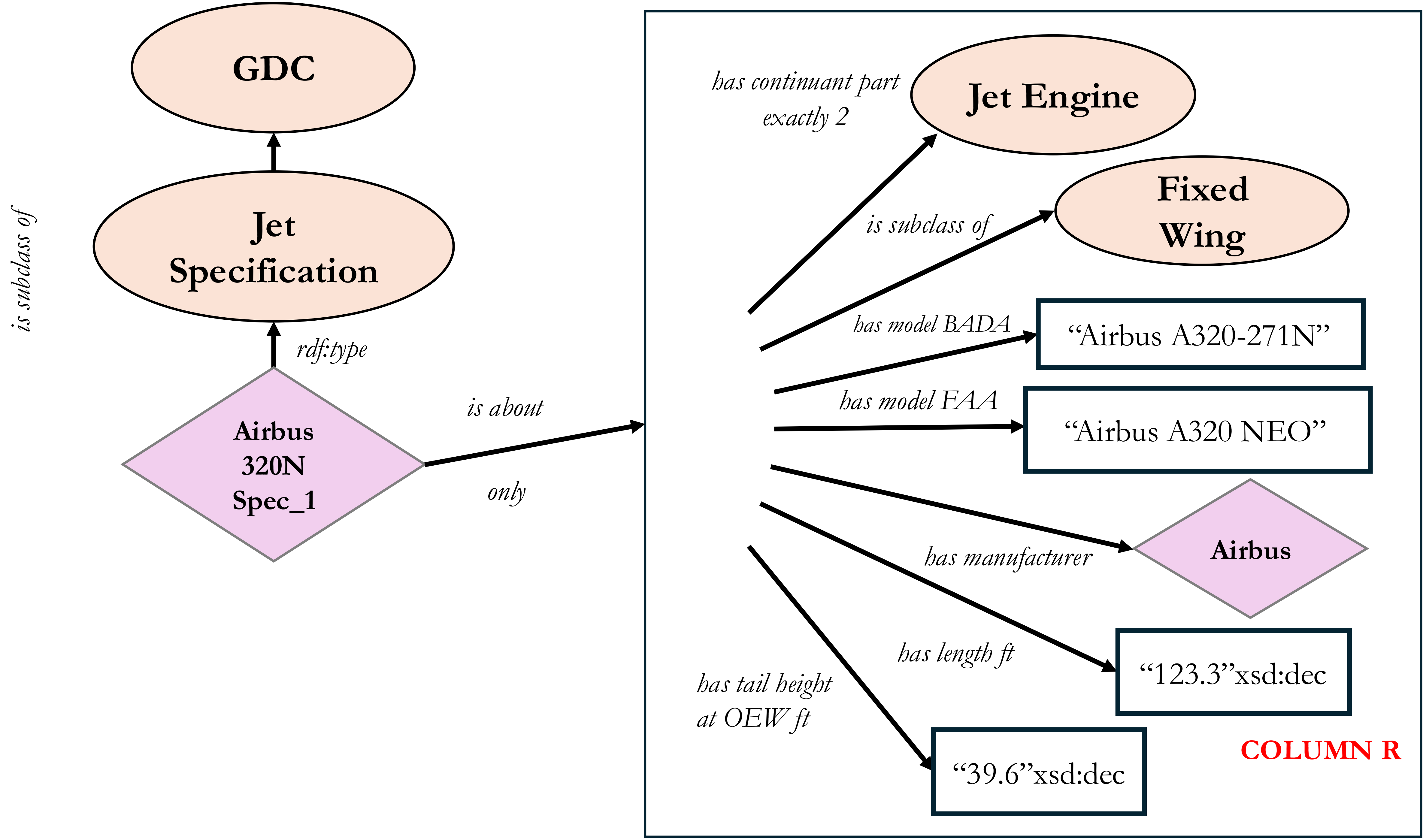
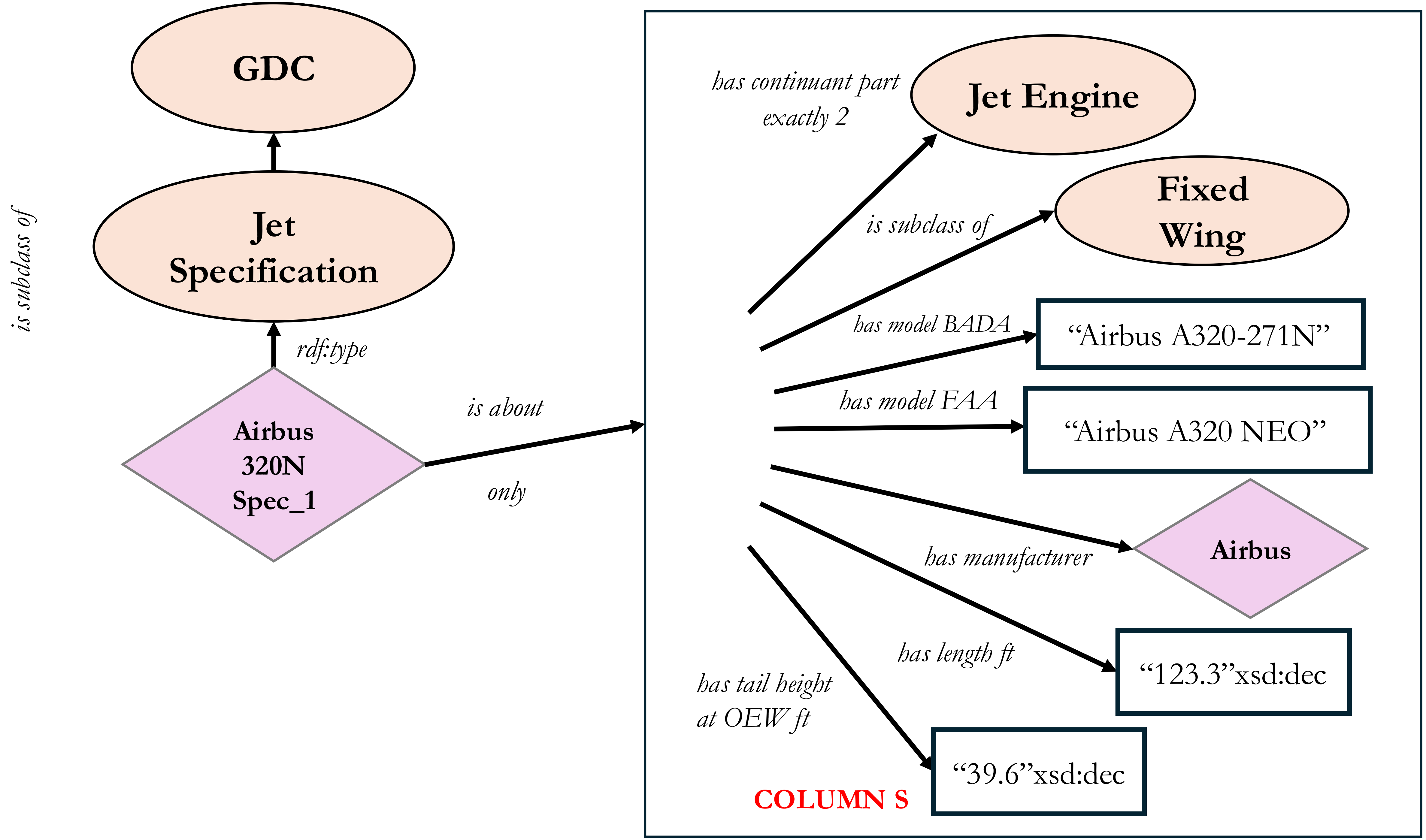
has tail height at OEW ft
"39.6"xsd:dec

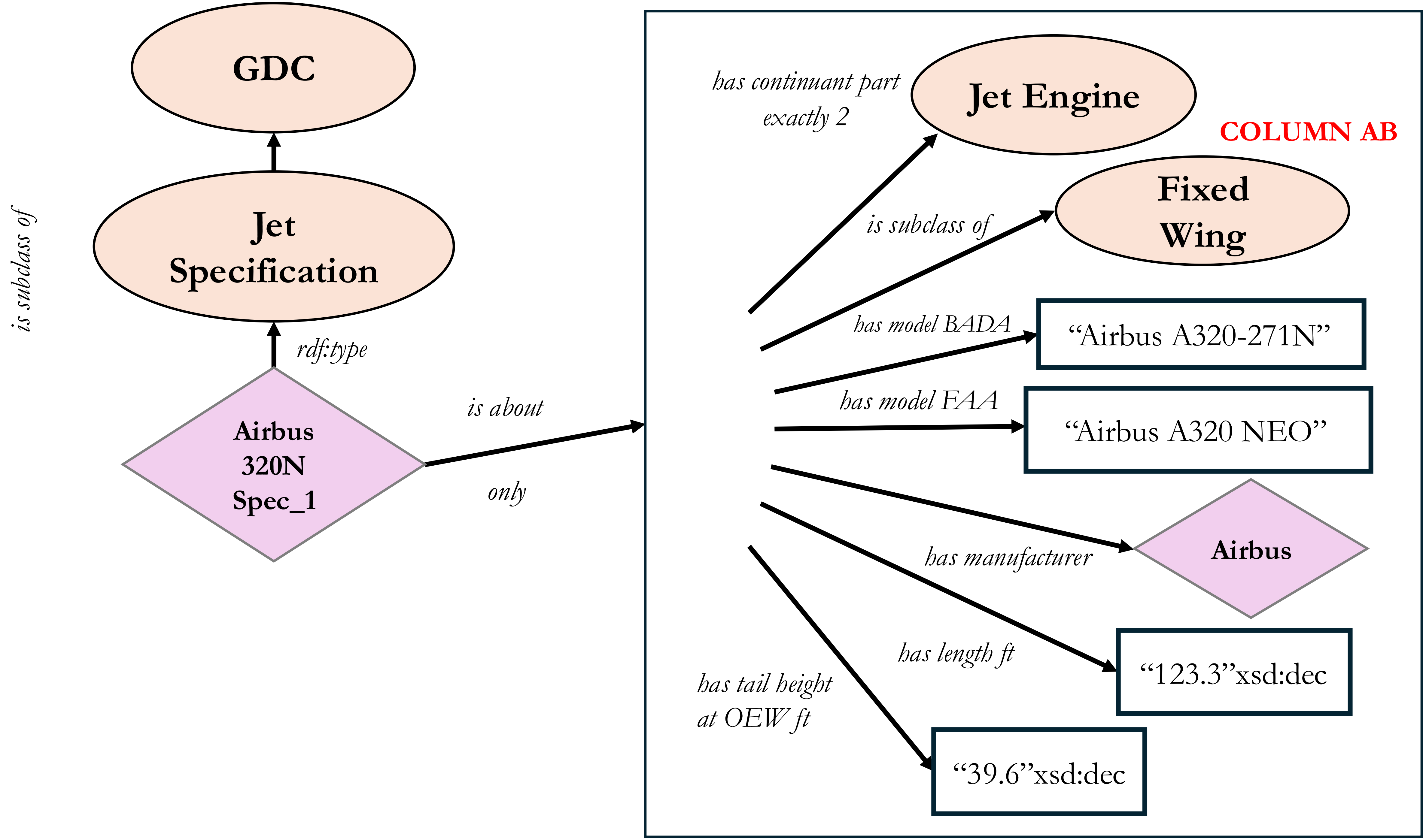


GDC
Jet Specification
is subclass of
rdf:type
Airbus 320N Spec_1
is about
only
COLUMN F
has continuant part exactly 2
Jet Engine
is subclass of
Fixed Wing
has model BADA
"Airbus A320-271N"
has model FAA
"Airbus A320 NEO"
has manufacturer
Airbus
has length ft
"123.3"xsd:dec
has tail height at OEW ft
"39.6"xsd:dec


GDC
Jet Specification
is subclass of
rdf:type
Airbus 320N Spec_1
is about
only
has continuant part exactly 2
COLUMN G
Jet Engine
is subclass of
Fixed Wing
has model BADA
"Airbus A320-271N"
has model FAA
"Airbus A320 NEO"
has manufacturer
Airbus
has length ft
"123.3"xsd:dec
has tail height at OEW ft
"39.6"xsd:dec


GDC
Jet Specification
rdf:type
Airbus 320N Spec_1
is subclass of
is about
only
has continuant part exactly 2
Jet Engine
is subclass of
Fixed Wing
has model BADA
"Airbus A320-271N"
has model FAA
"Airbus A320 NEO"
has manufacturer
Airbus
has length ft
"123.3"xsd:dec
has tail height at OEW ft
"39.6"xsd:dec
COLUMN R


GDC
Jet Specification
is subclass of
rdf:type
Airbus 320N Spec_1
is about
only
has continuant part exactly 2
Jet Engine
is subclass of
Fixed Wing
has model BADA
"Airbus A320-271N"
has model FAA
"Airbus A320 NEO"
has manufacturer
Airbus
has length ft
"123.3"xsd:dec
has tail height at OEW ft
"39.6"xsd:dec
COLUMN S


GDC
Jet Specification
rdf:type
Airbus 320N Spec_1
is about
only
is subclass of
has continuant part exactly 2
Jet Engine
COLUMN AB
is subclass of
Fixed Wing
has model BADA
"Airbus A320-271N"
has model FAA
"Airbus A320 NEO"
has manufacturer
Airbus
has length ft
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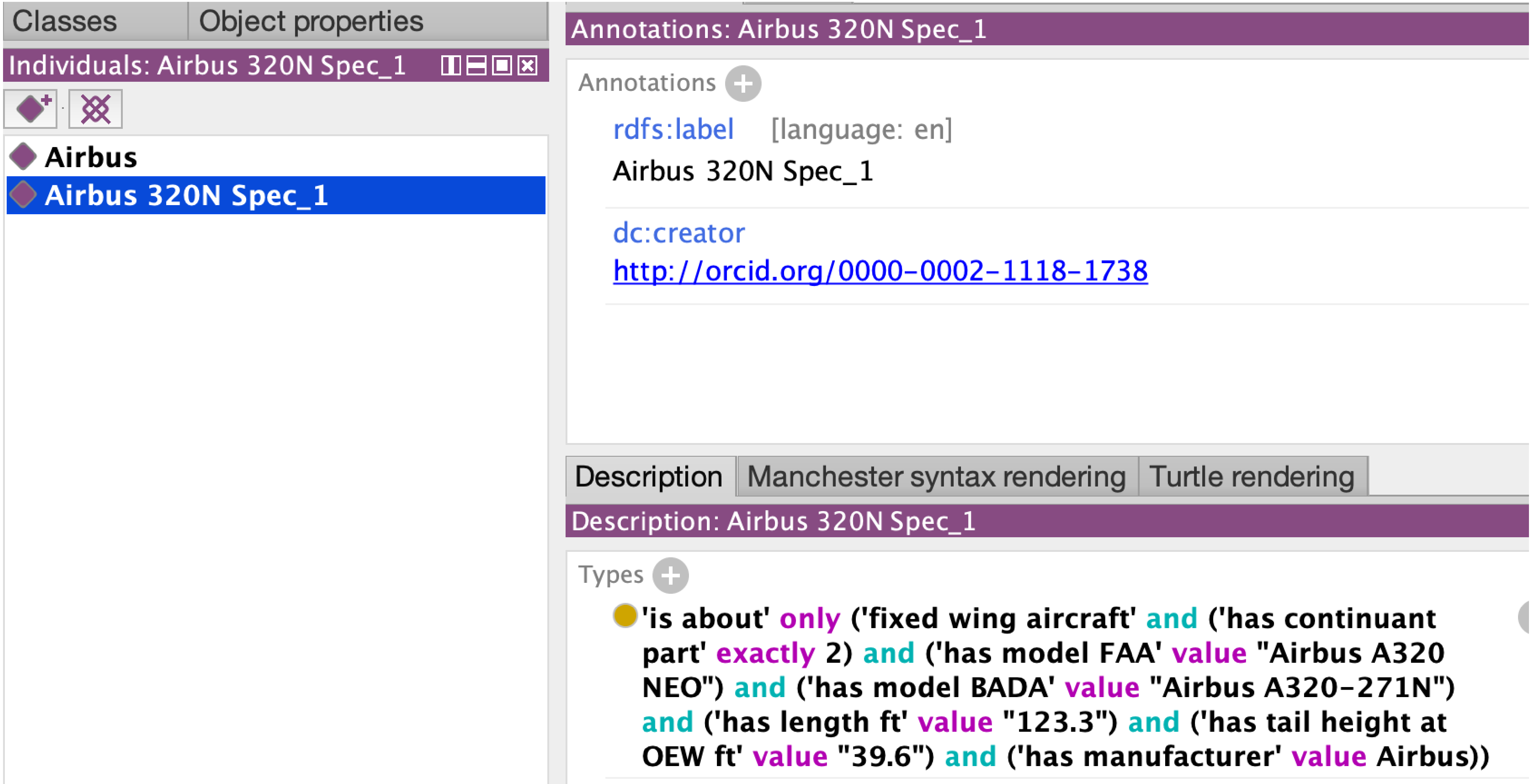
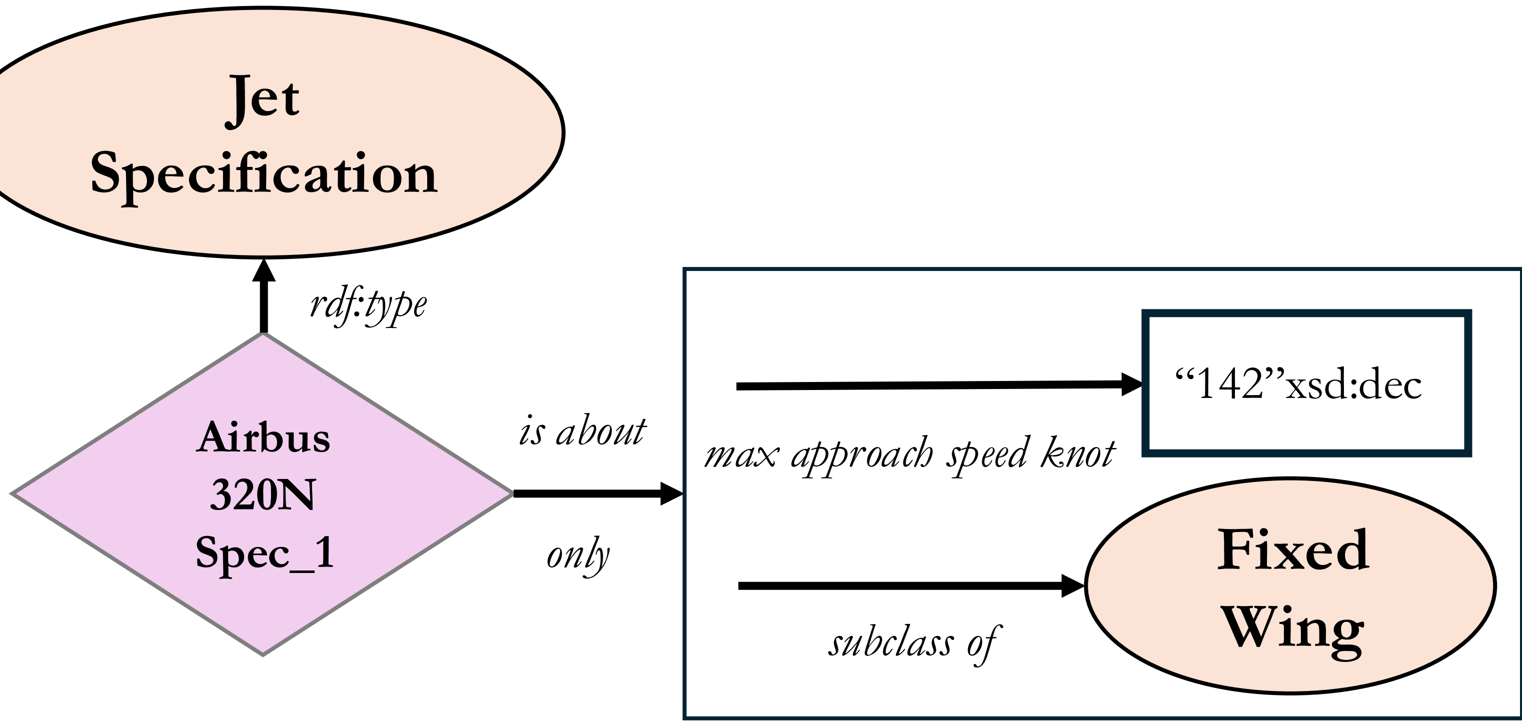
has tail height at OEW ft
"39.6"xsd:dec


Classes
Object properties
Individuals: Airbus 320N Spec_1
Airbus
Airbus 320N Spec_1
Annotations: Airbus 320N Spec_1
Annotations
rdfs:label [language: en]
Airbus 320N Spec_1
dc:creator
http://orcid.org/0000-0002-1118-1738
Description
Manchester syntax rendering
Turtle rendering
Description: Airbus 320N Spec_1
Types
'is about' only ('fixed wing aircraft' and ('has continuant part' exactly 2) and ('has model FAA' value "Airbus A320 NEO") and ('has model BADA' value "Airbus A320-271N") and ('has length ft' value "123.3") and ('has tail height at OEW ft' value "39.6") and ('has manufacturer' value Airbus))

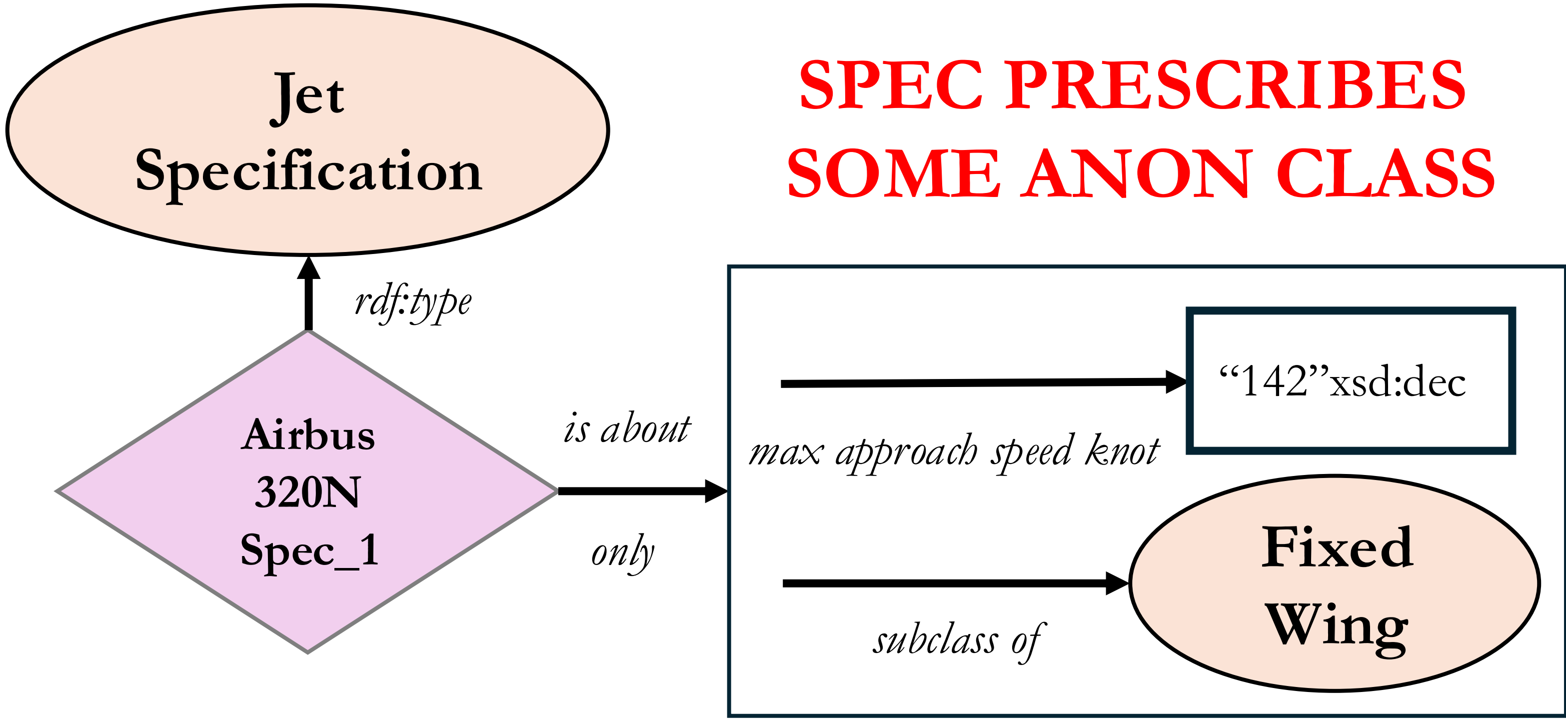
Case 2

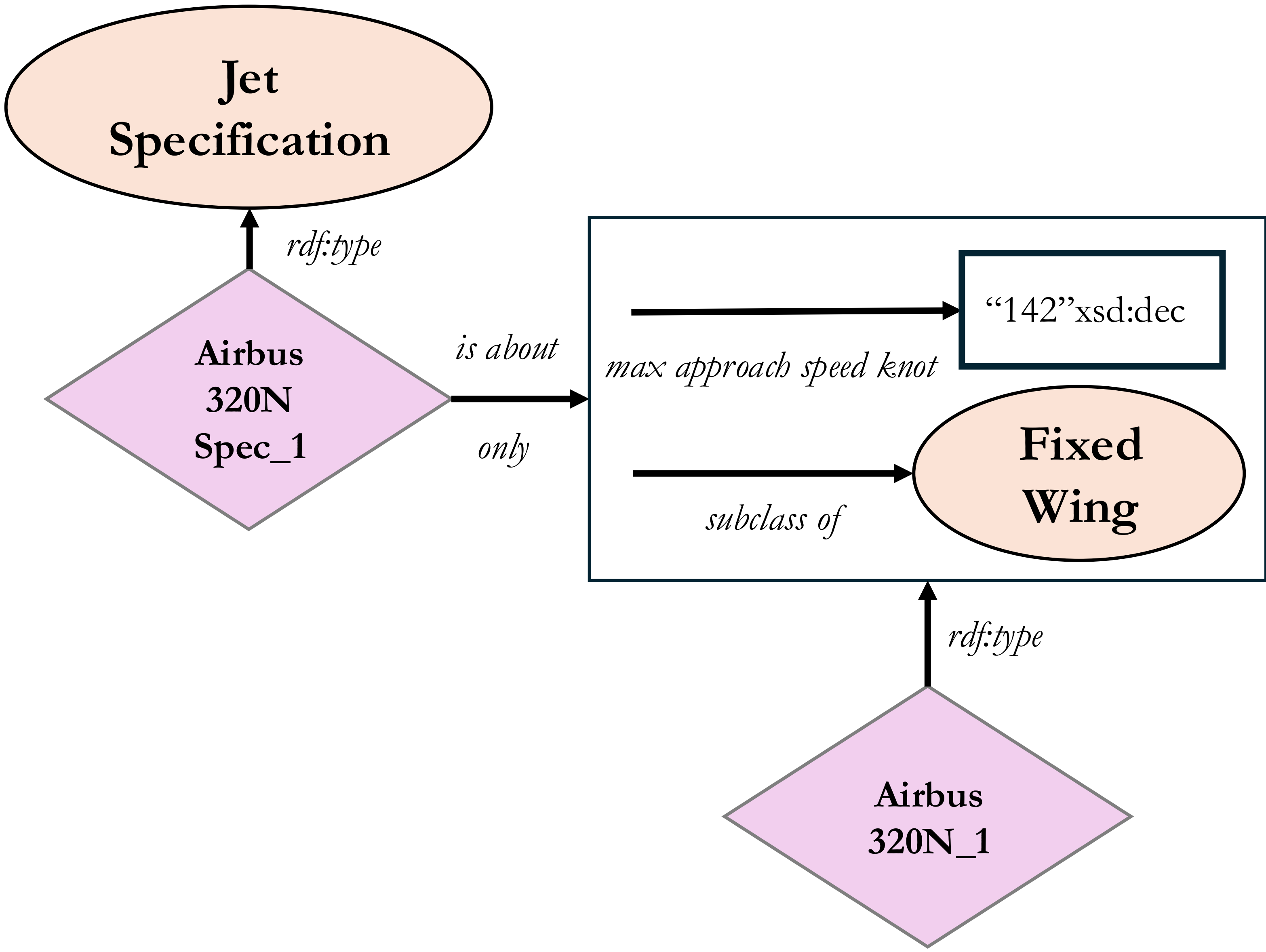
In aircraft_data.xlsx you will find a row for the Airbus A321-111, designed to have a maximum knot approach speed of 142. However, after 5 approaches, an instance has obtained an average maximum knot approach speed of 139.

Construct a design pattern and OWL encoding reflecting the preceding phenomena.

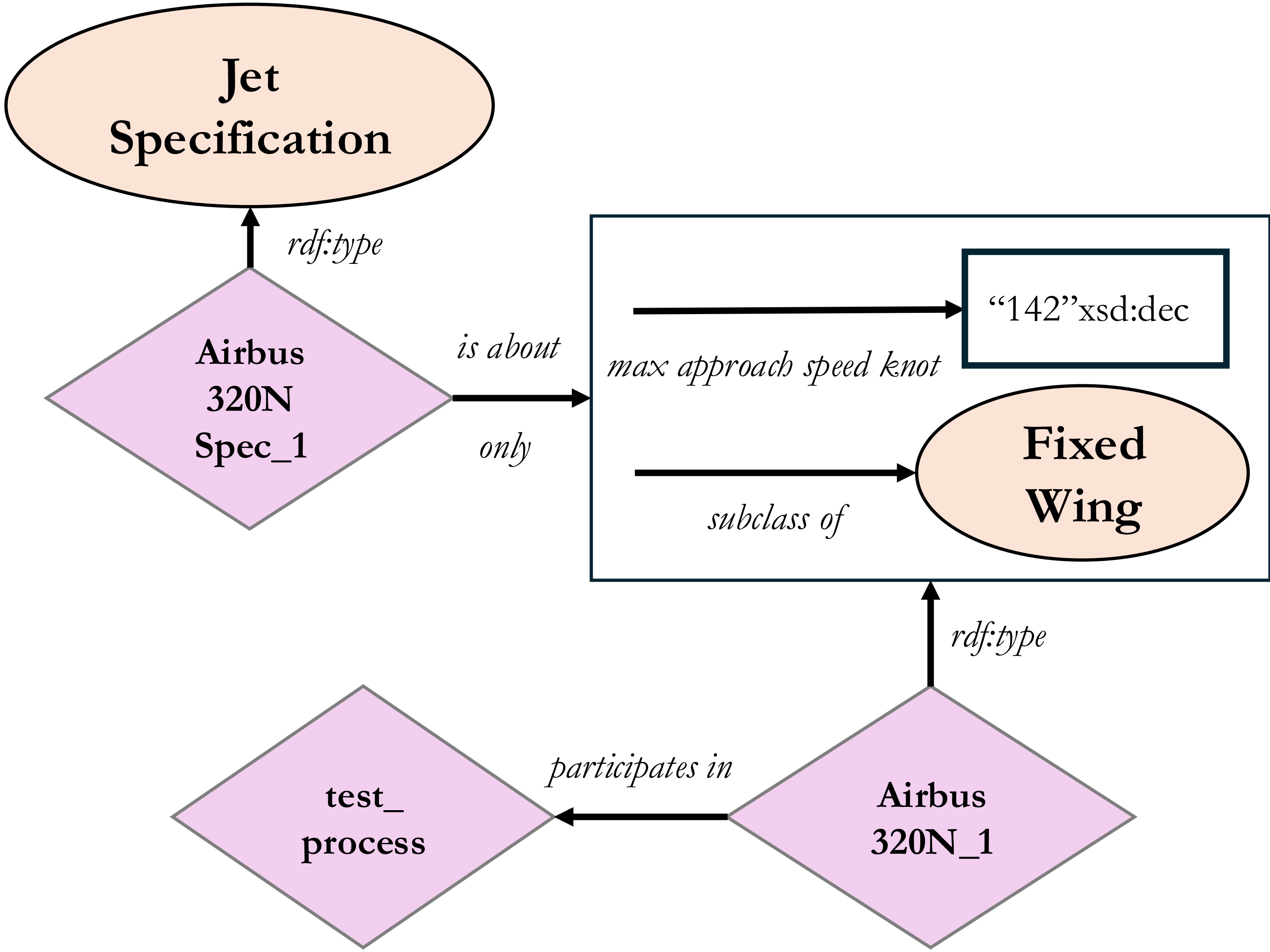

Jet Specification
rdf:type
Airbus 320N Spec_1
is about
only
max approach speed knot
“142”xsd:dec
subclass of
Fixed Wing

SPEC PRESCRIBES SOME ANON CLASS

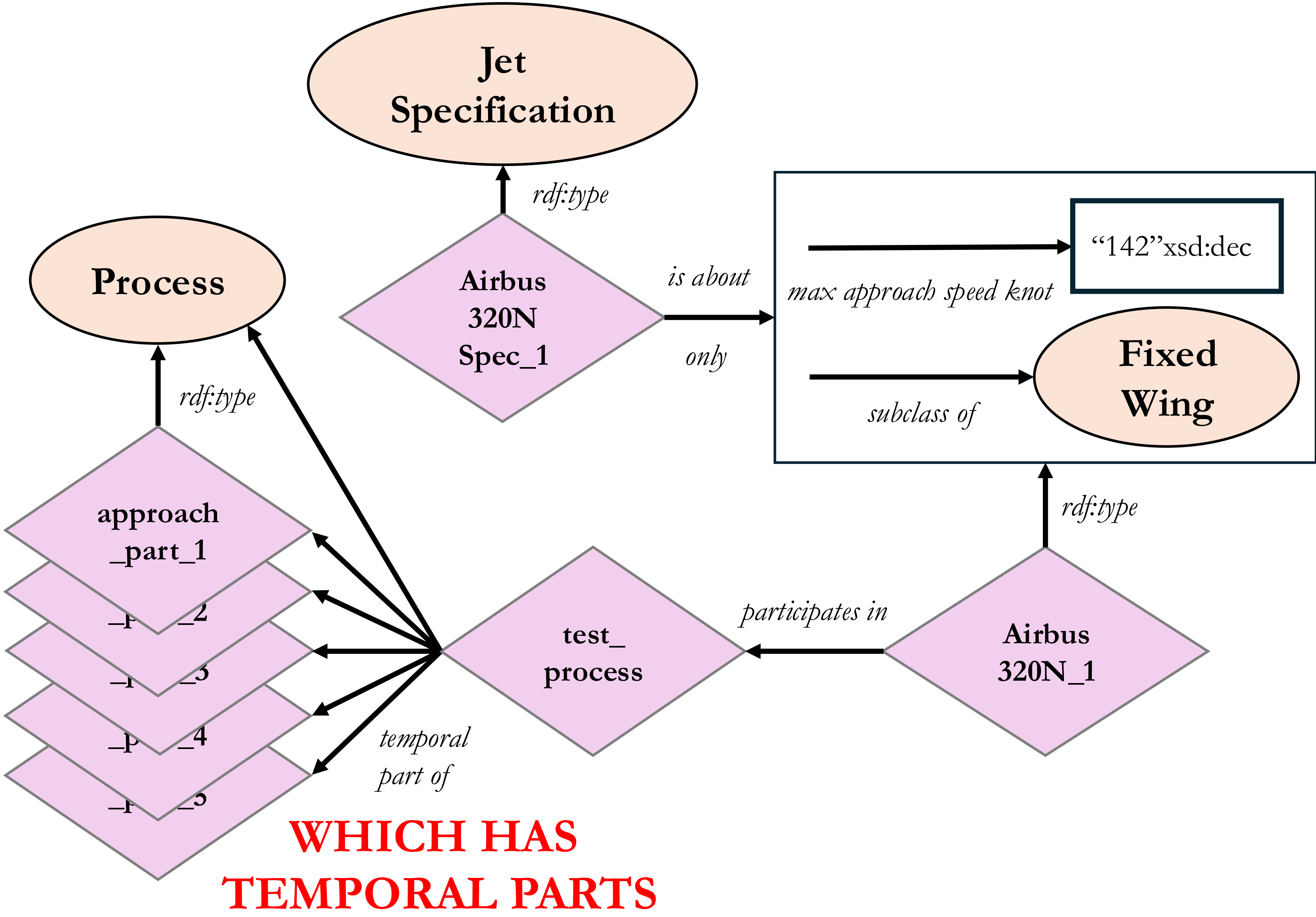


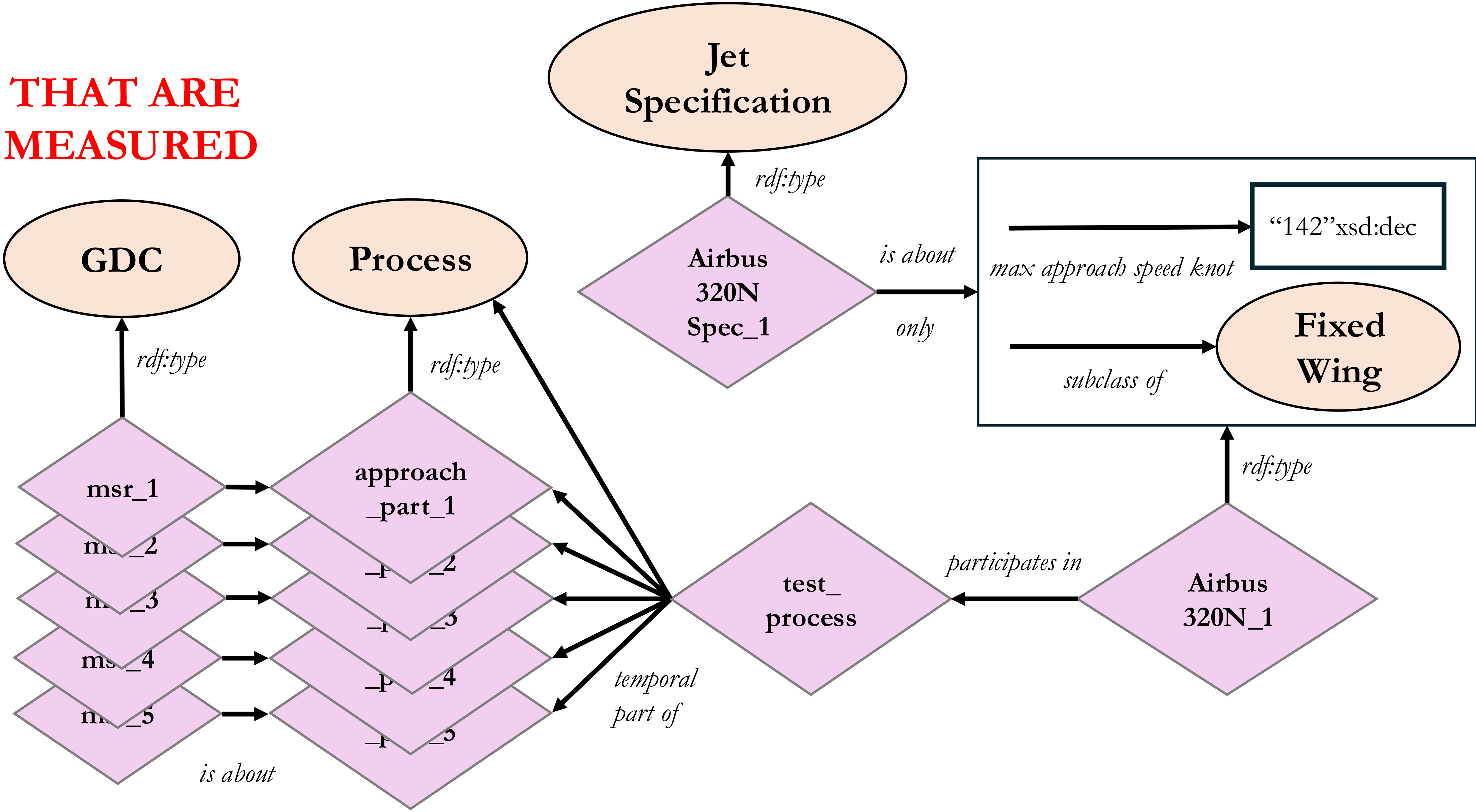


WHICH HAS A FIXED WING INSTANCE

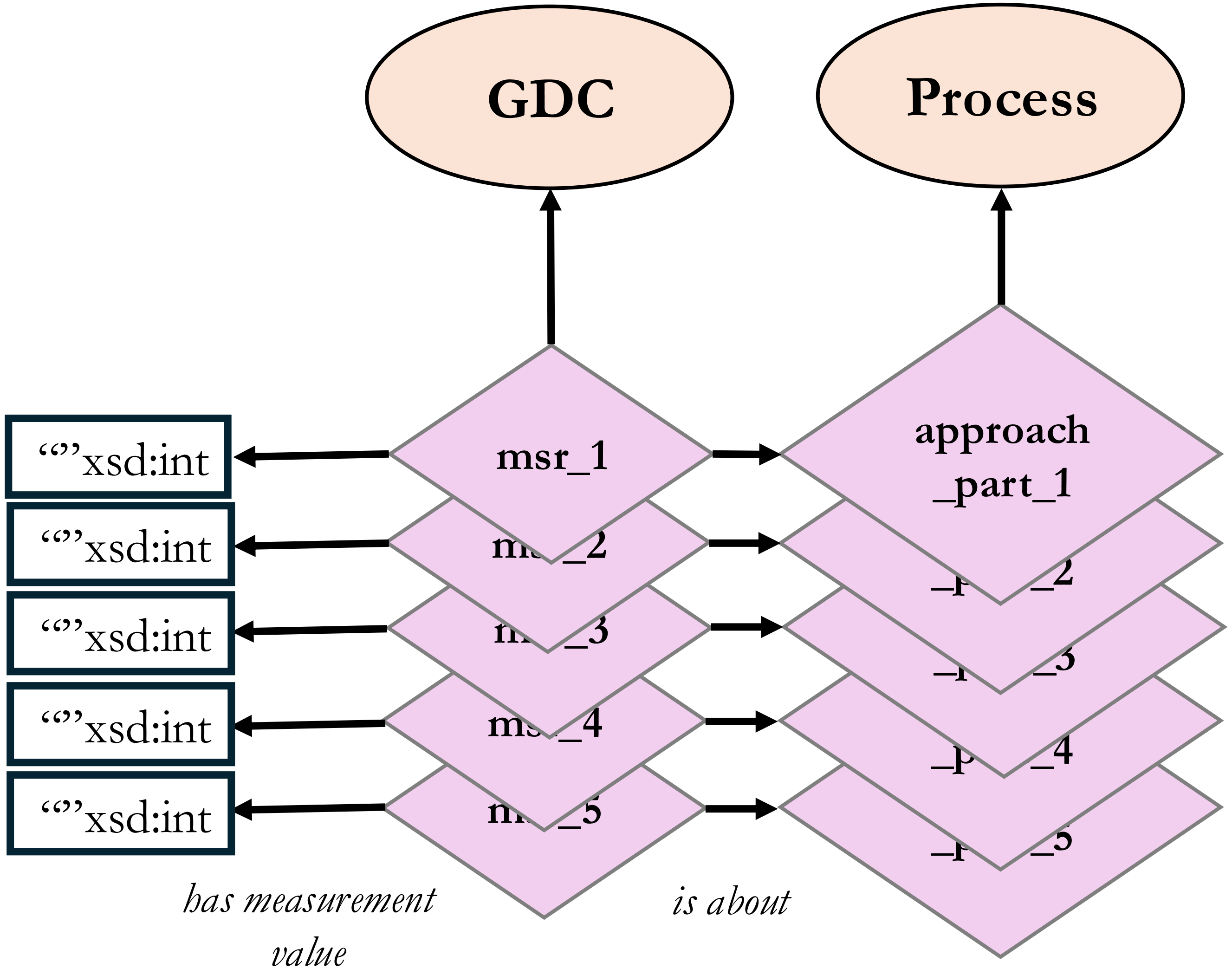


THAT PARTICIPATES IN
AN OVERALL TEST PROCESS

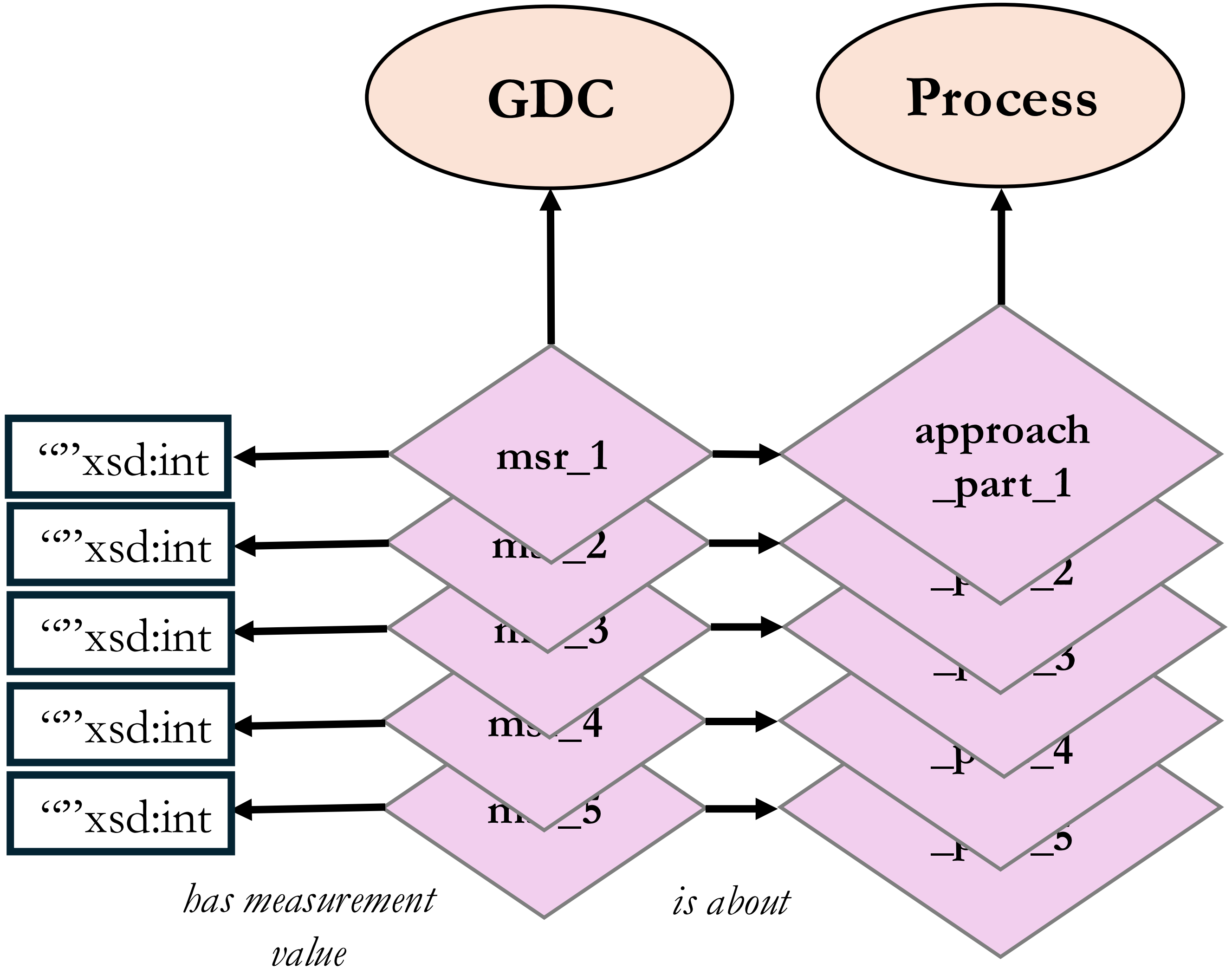

Jet Specification
rdf:type
Process
Airbus 320N Spec_1
is about
only
"142"xsd:dec
max approach speed knot
Fixed Wing
subclass of
rdf:type
approach _part_1
rdf:type
test_ process
participates in
Airbus 320N_1
temporal part of
WHICH HAS TEMPORAL PARTS

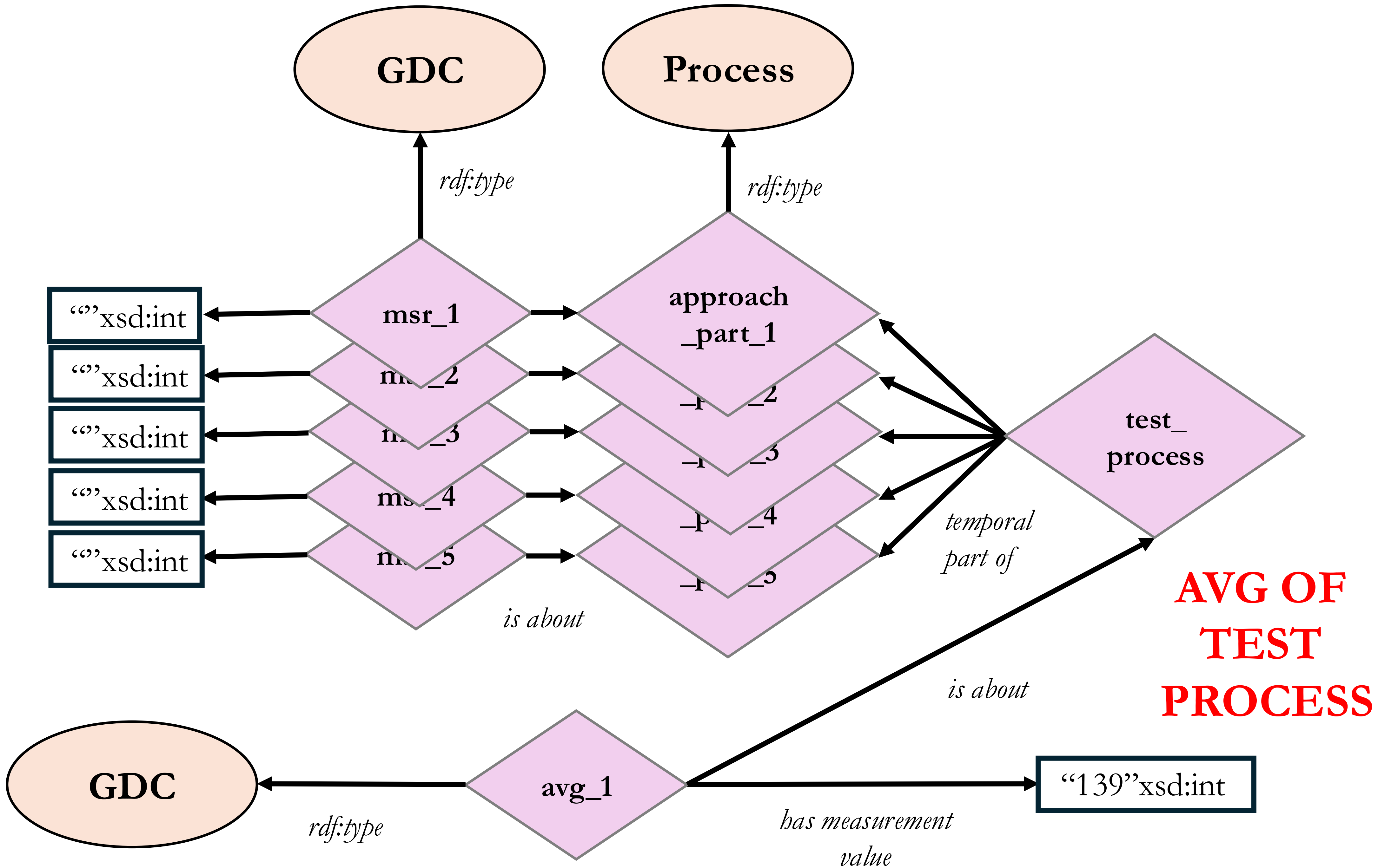

THAT ARE MEASURED
Jet Specification
rdf:type
Airbus 320N Spec_1
is about
only
max approach speed knot
"142"xsd:dec
subclass of
Fixed Wing
rdf:type
GDC
Process
rdf:type
rdf:type
msr_1
approach _part_1
test_ process
participates in
Airbus 320N_1
temporal part of
is about

SUPPOSE WE HAVE MEASUREMENT VALUES



USE SPARQL TO CALCULATE AND RETURN THE AVERAGE





GDC
Process
rdf:type
rdf:type
msr_1
approach
_part_1
“”xsd:int
“”xsd:int
“”xsd:int
“”xsd:int
“”xsd:int
_2
_3
_4
_5
test_
process
temporal
part of
is about
AVG OF
TEST
PROCESS
is about
GDC
avg_1
“139”xsd:int
rdf:type
has measurement
value

Case 3

In soc_structure_definitions.xlsx you will find three "SOC_TITLE" entries that mention "Aerospace Engineer".

Construct a design pattern and OWL encoding that reflects all three entries and their respective "SOC Definitions".

Major Group	Minor Group	Broad Group	Detailed Occupation
17-2000			**Engineers**
	17-2010		Aerospace Engineers
		17-2011	Aerospace Engineers
		17-3021	Aerospace Engineering and Operations Technologists and Technicians

US SOC Data Structure

Level	Digits	Example	Description
Major Group	2 digits	15-0000	Very broad category (e.g., Computer and Mathematical Occupations)
Minor Group	4 digits	15-1200	Subcategory under major group (e.g., Computer Occupations)
Broad Occupation	5 digits	15-1250	A grouping of related detailed occupations (e.g., Software and Web Developers)
Detailed Occupation	6 digits	15-1252	A specific job type (e.g., Software Developers)

US SOC Data Structure

Level	Digits	Example	Description
Major Group	2 digits	15-0000	Very broad category (e.g., Computer and Mathematical Occupations)
Minor Group	4 digits	15-1200	Subcategory under major group (e.g., Computer Occupations)
Broad Occupation	5 digits	15-1250	A grouping of related detailed occupations (e.g., Software and Web Developers)
Detailed Occupation	6 digits	15-1252	A specific job type (e.g., Software Developers)

Broad & Detailed

- You will need to learn to deal with **confusing data**:

17-2010 Aerospace Engineers

This broad occupation is the same as the detailed occupation:

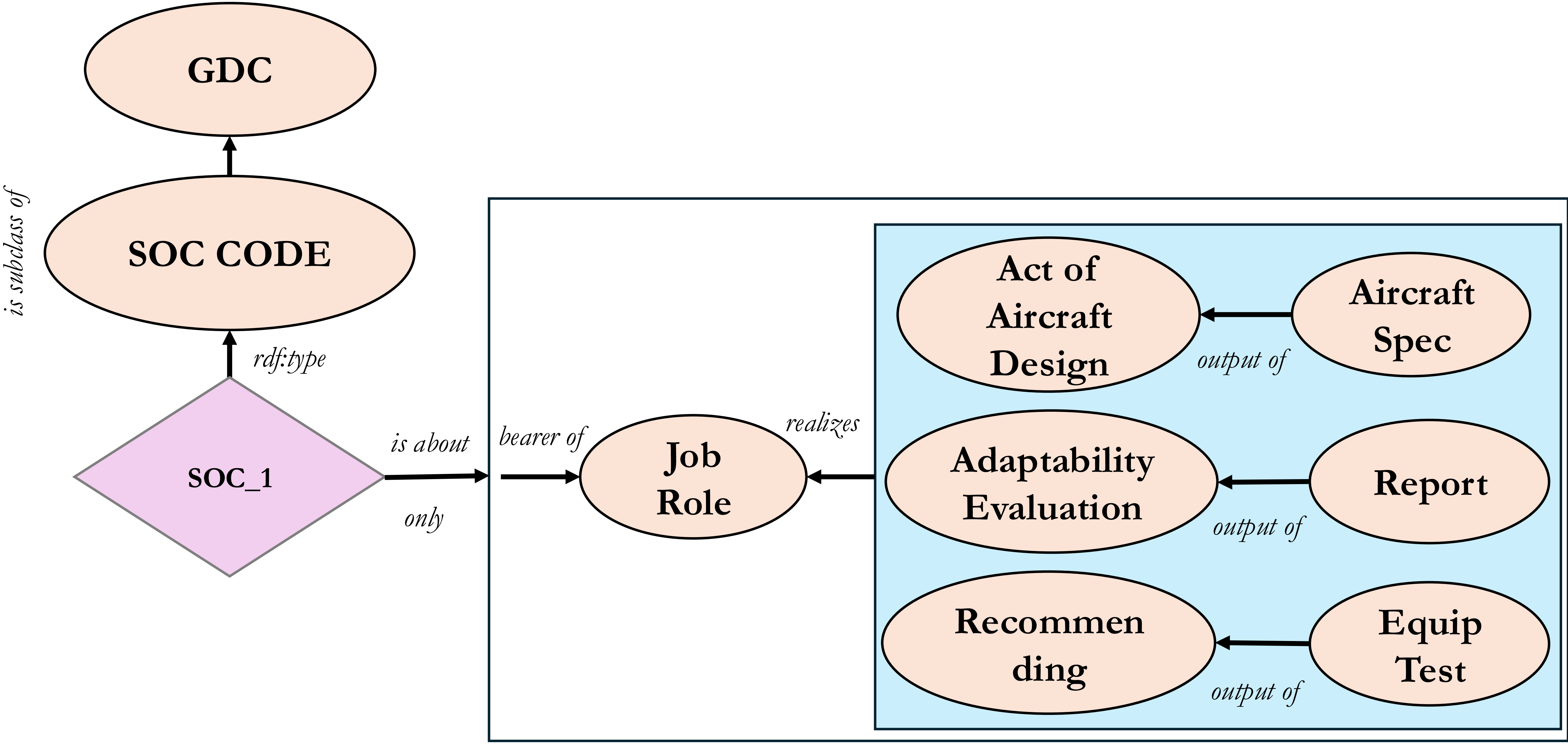
17-2011 Aerospace Engineers

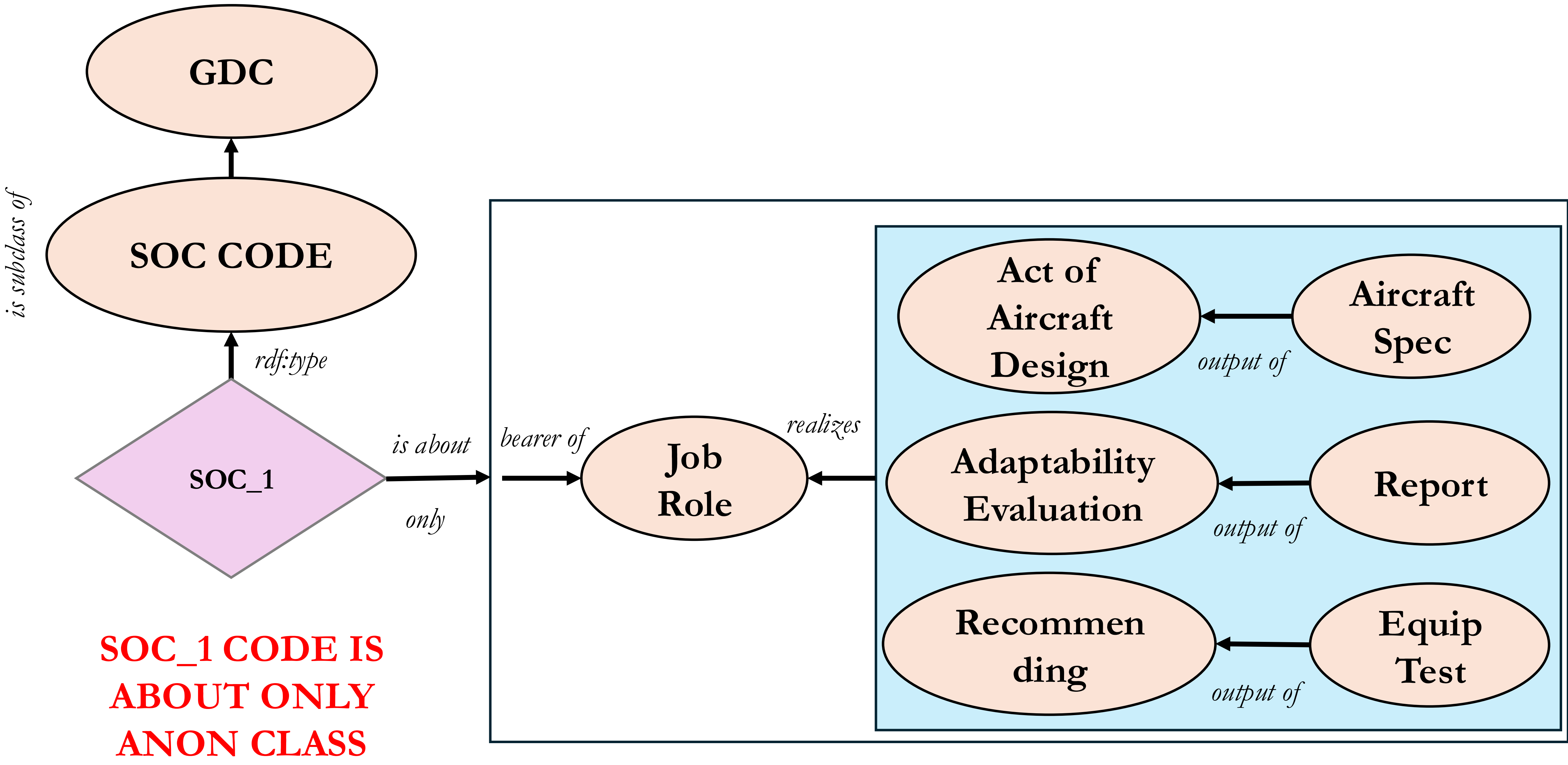
- **Always** research:

https://www.bls.gov/soc/2018/soc_2018_manual.pdf

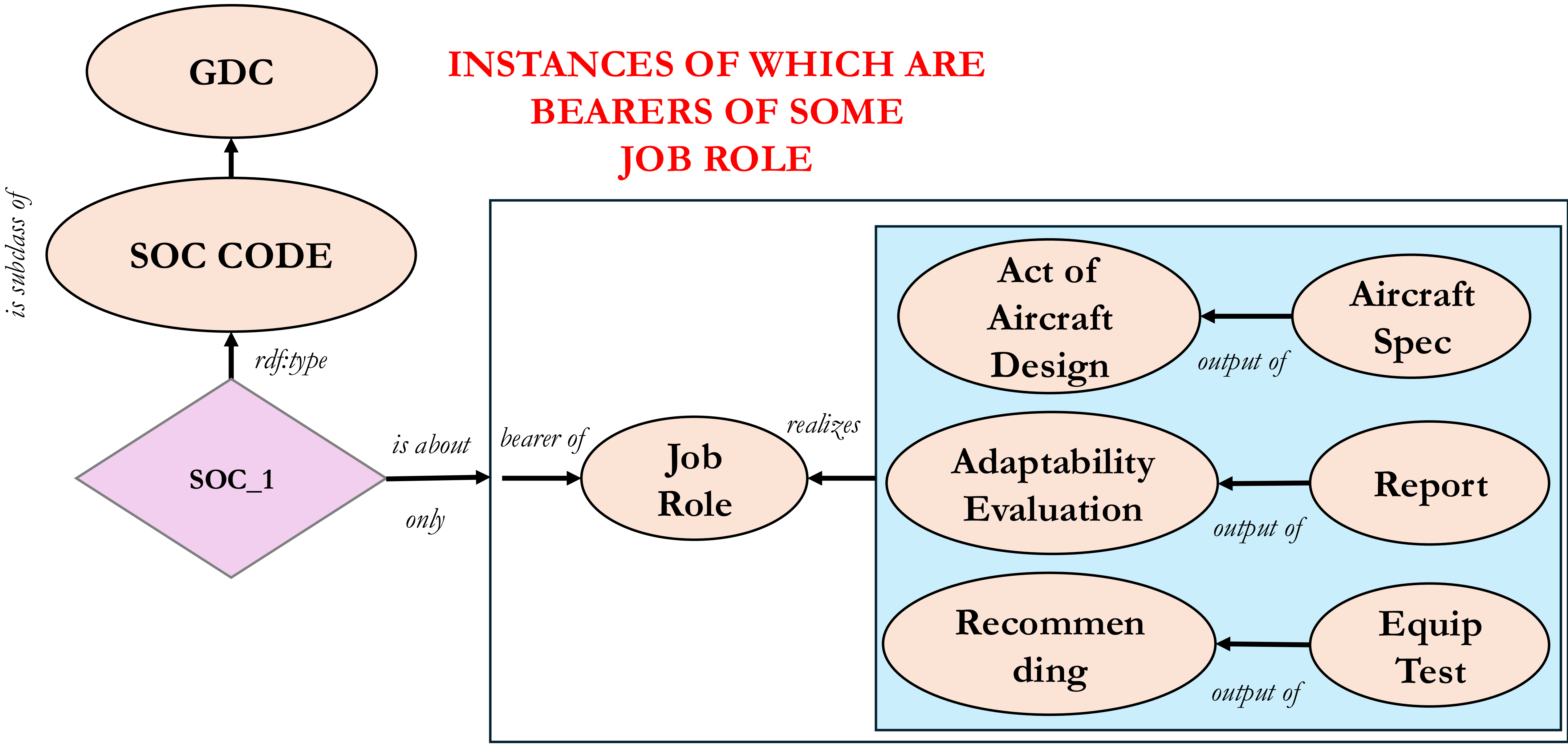
Aerospace Engineer

Perform engineering duties in designing, constructing, and testing aircraft, missiles, and spacecraft. May conduct basic and applied research to evaluate adaptability of materials and equipment to aircraft design and manufacture. May recommend improvements in testing equipment and techniques.

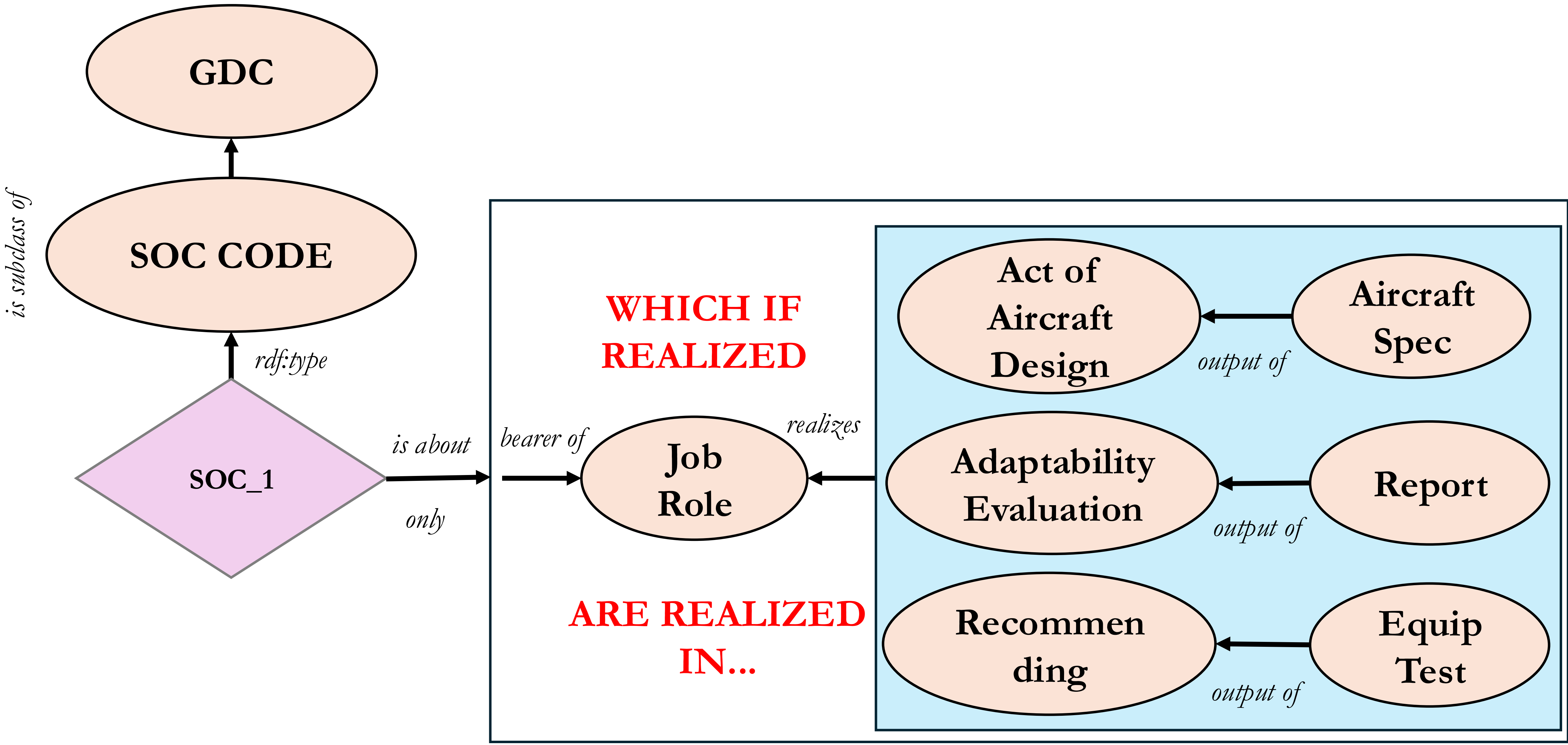

GDC
is subclass of
SOC CODE
rdf:type
SOC_1
is about
only
bearer of
Job Role
realizes
Act of Aircraft Design
output of
Aircraft Spec
Adaptability Evaluation
output of
Report
Recommending
output of
Equip Test

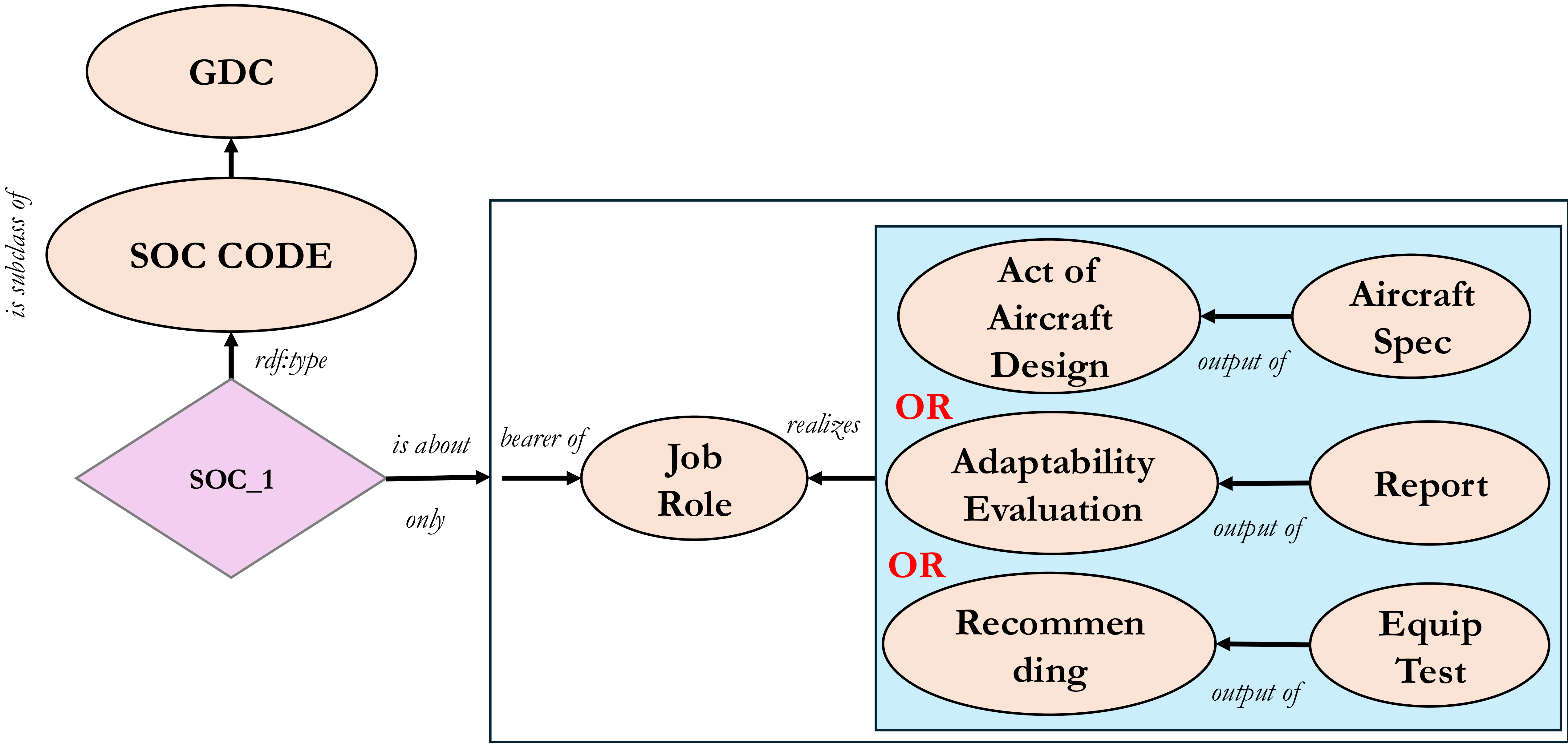


GDC
is subclass of
SOC CODE
rdf:type
SOC_1
is about
only
bearer of
Job Role
realizes
Act of Aircraft Design
output of
Aircraft Spec
Adaptability Evaluation
output of
Report
Recommending
output of
Equip Test
SOC_1 CODE IS ABOUT ONLY ANON CLASS



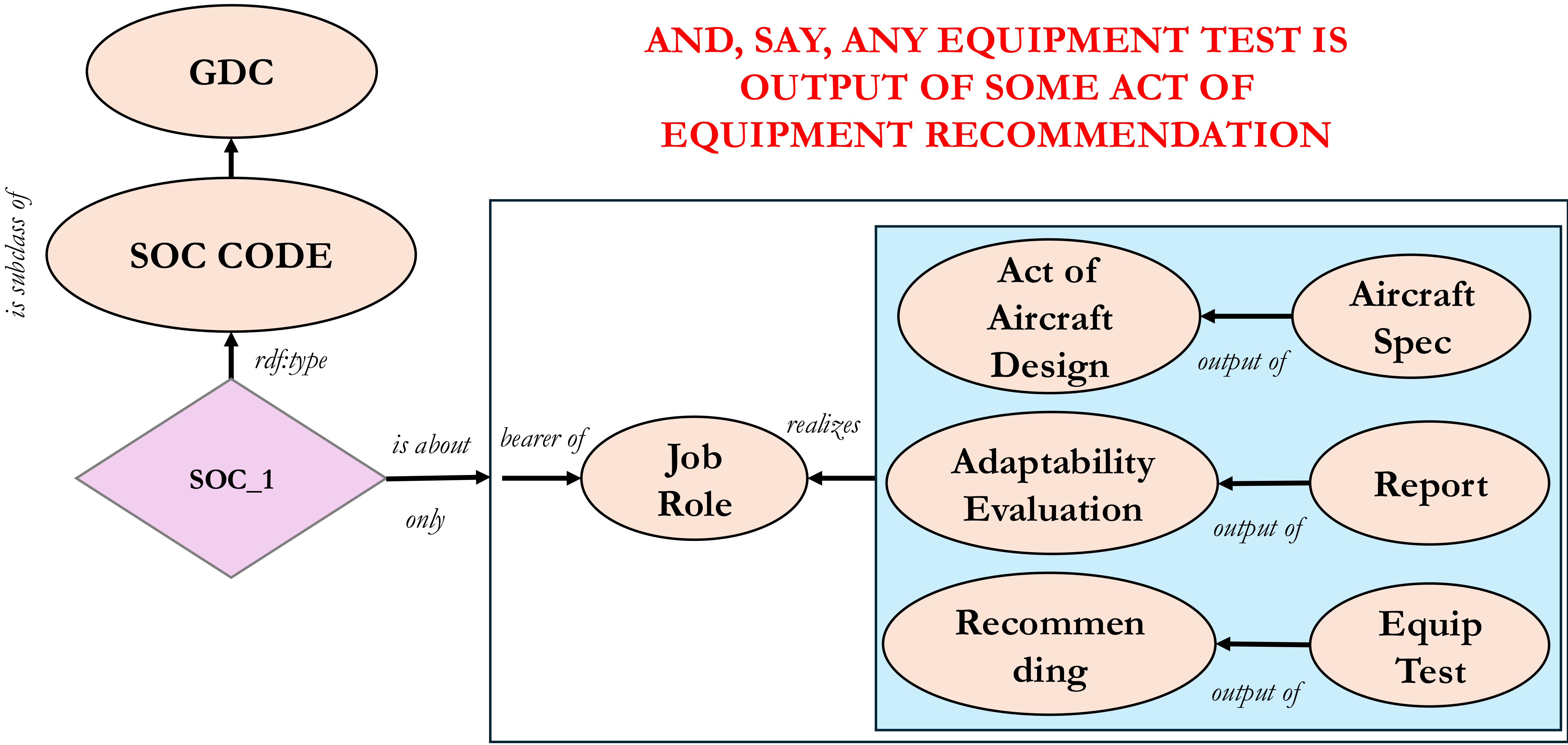
INSTANCES OF WHICH ARE
BEARERS OF SOME
JOB ROLE
GDC
is subclass of
SOC CODE
rdf:type
SOC_1
is about
only
bearer of
Job Role
realizes
Act of Aircraft Design
output of
Aircraft Spec
Adaptability Evaluation
output of
Report
Recommending
output of
Equip Test


GDC
is subclass of
SOC CODE
rdf:type
SOC_1
is about
only
bearer of
WHICH IF
REALIZED
Job
Role
realizes
ARE REALIZED
IN...
Act of
Aircraft
Design
output of
Aircraft
Spec
Adaptability
Evaluation
output of
Report
Recommen
ding
output of
Equip
Test



GDC
is subclass of
SOC CODE
rdf:type
SOC_1
is about
only
bearer of
Job Role
realizes
Act of Aircraft Design
output of
Aircraft Spec
OR
Adaptability Evaluation
output of
Report
OR
Recommending
output of
Equip Test

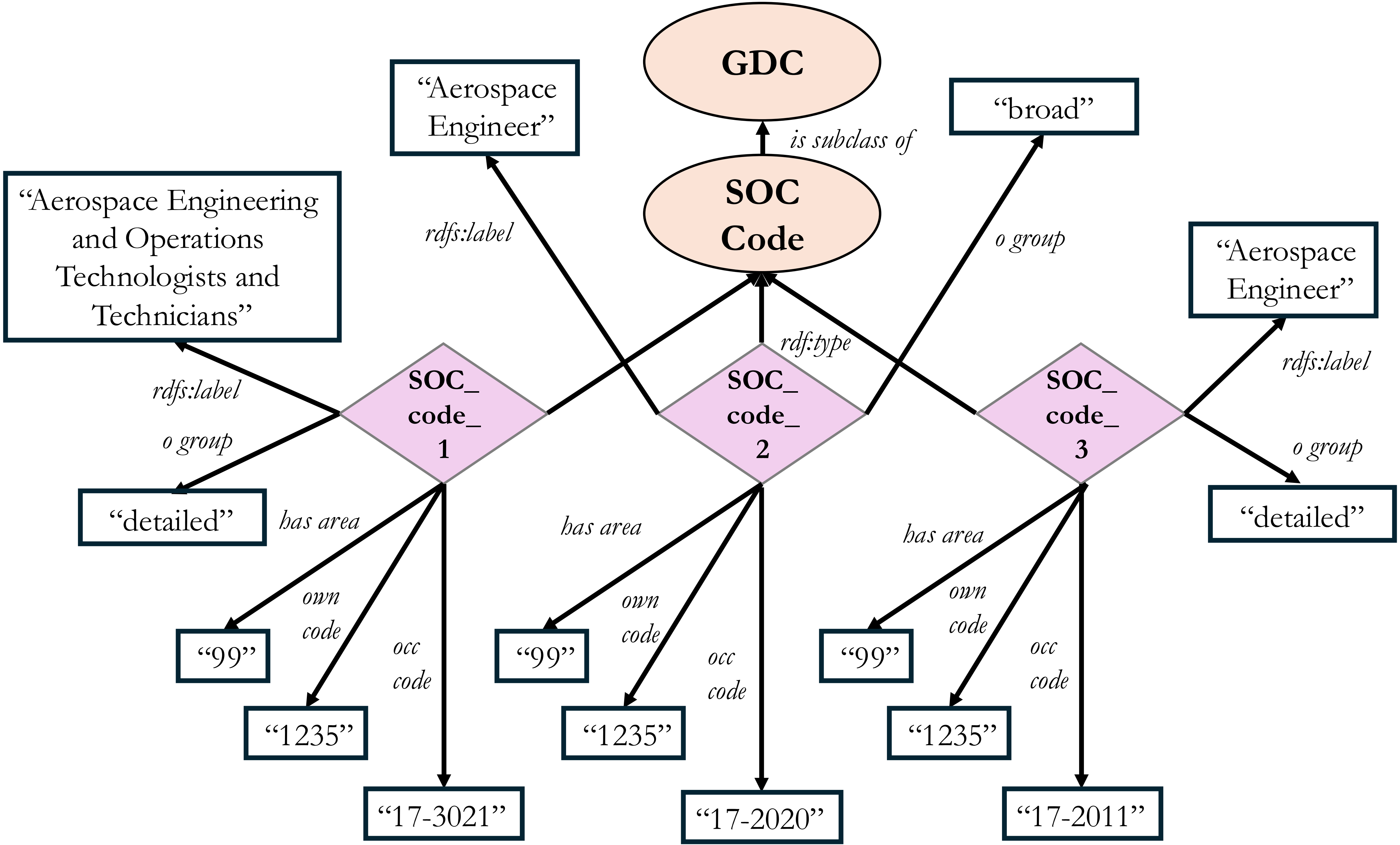
AND, SAY, ANY EQUIPMENT TEST IS OUTPUT OF SOME ACT OF EQUIPMENT RECOMMENDATION

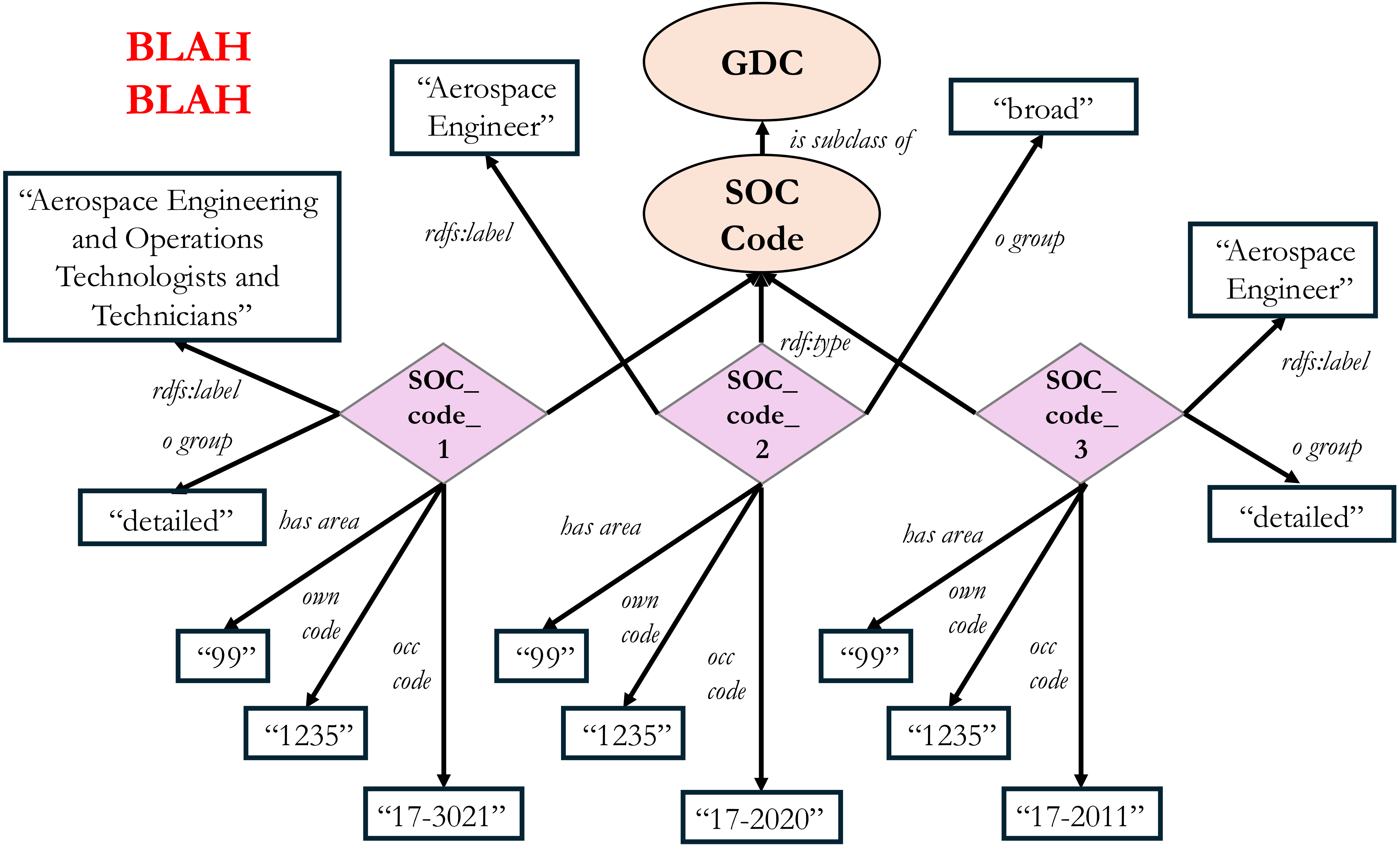


Case 4

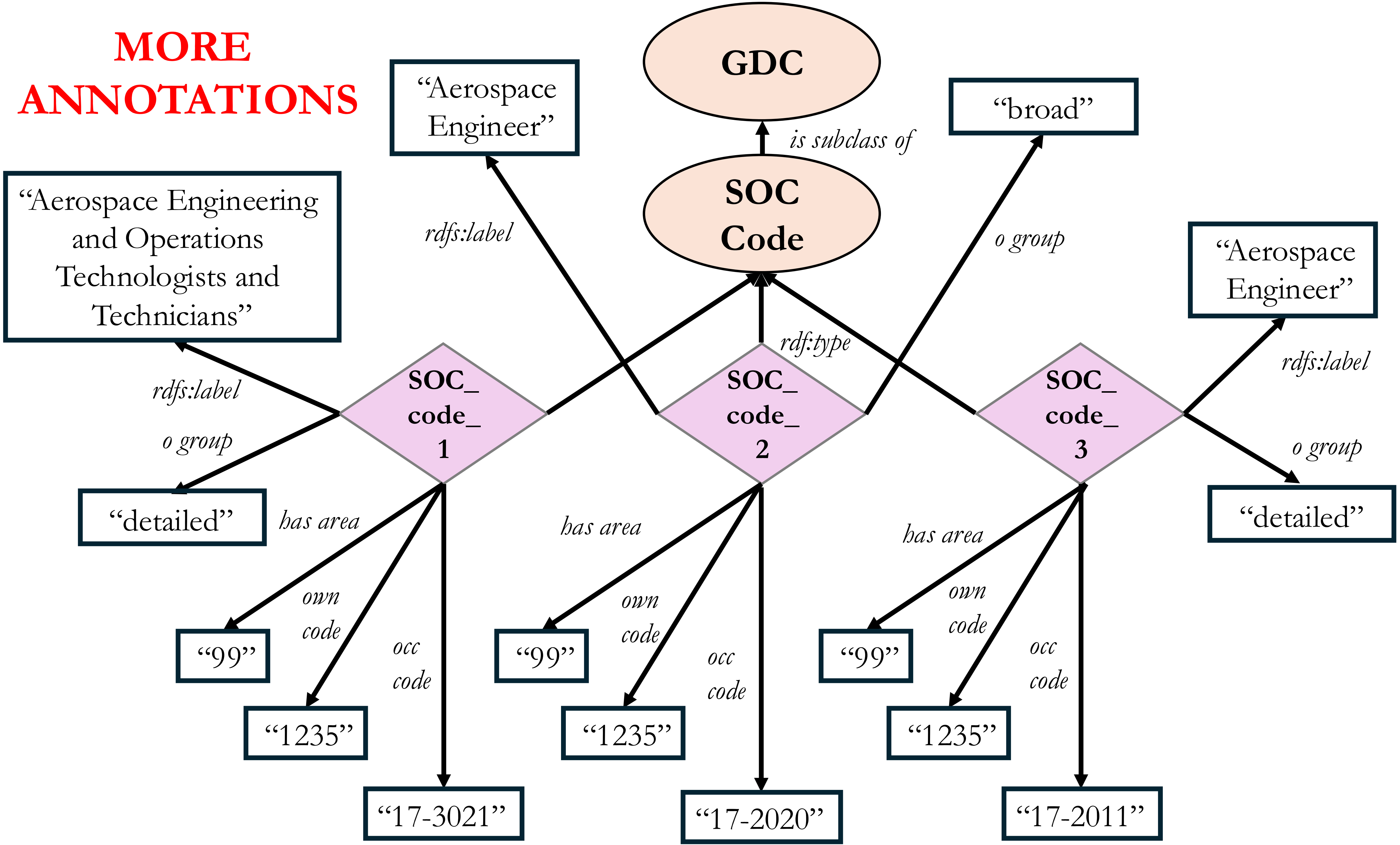
In employment_wage_May_2024.xlsx you will find three "OCC_TITLE" entries that mention "Aerospace Engineer".

Construct a design pattern and OWL encoding that reflects all three entries and their associated columns A, H, I, K, L.


GDC
is subclass of
SOC Code
"Aerospace Engineer"
"broad"
"Aerospace Engineering and Operations Technologists and Technicians"
"Aerospace Engineer"
rdfs:label
o group
rdf:type
rdfs:label
SOC_ code_ 1
SOC_ code_ 2
SOC_ code_ 3
rdfs:label
o group
o group
"detailed"
"detailed"
has area
has area
has area
own code
own code
own code
occ code
occ code
occ code
"99"
"99"
"99"
"1235"
"1235"
"1235"
"17-3021"
"17-2020"
"17-2011"

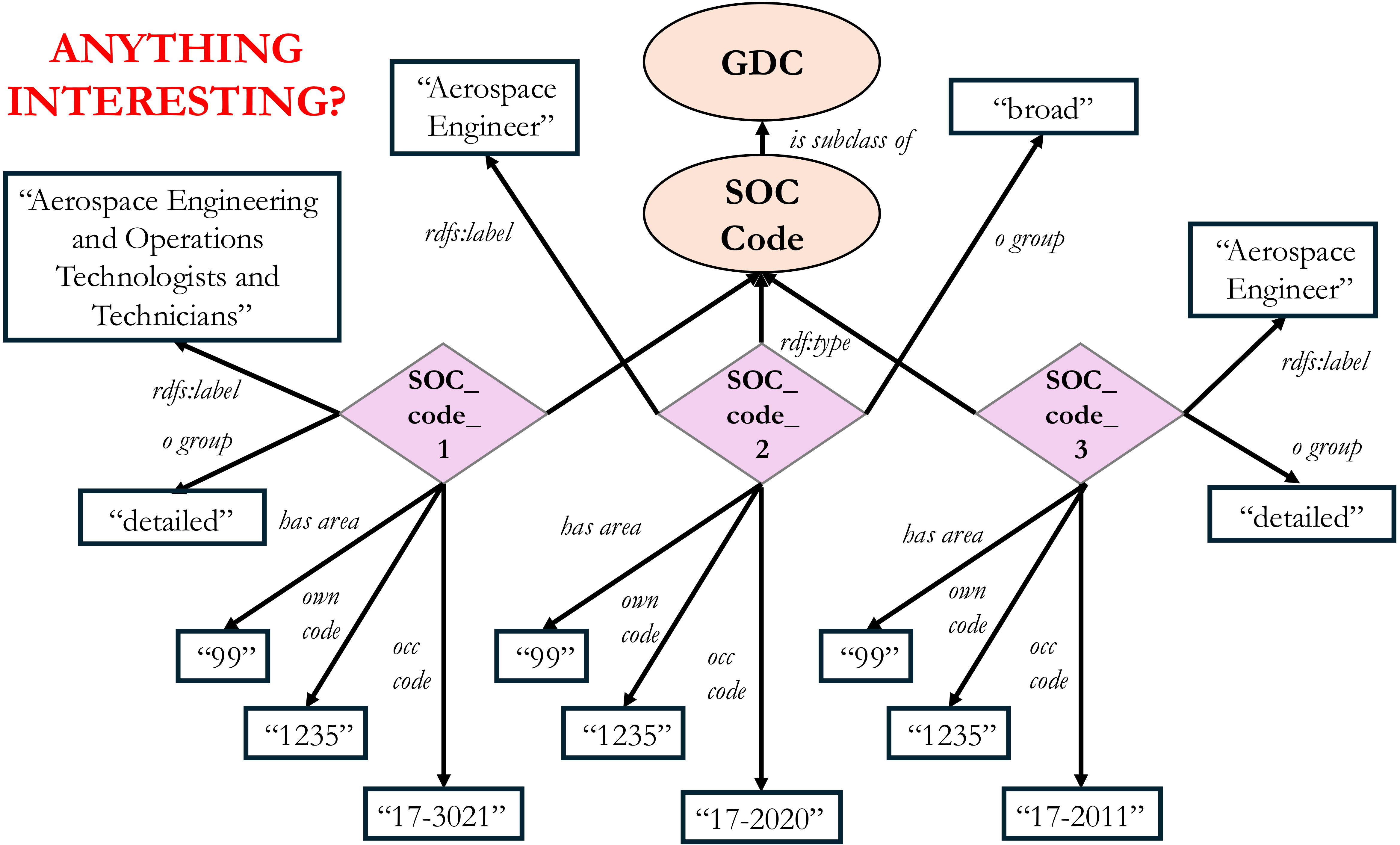


BLAH
BLAH
GDC
is subclass of
SOC Code
"Aerospace Engineer"
"broad"
"Aerospace Engineering and Operations Technologists and Technicians"
rdfs:label
o group
rdf:type
"Aerospace Engineer"
rdfs:label
SOC_ code_ 1
SOC_ code_ 2
SOC_ code_ 3
rdfs:label
o group
o group
"detailed"
"detailed"
has area
has area
has area
own code
own code
own code
occ code
occ code
occ code
"99"
"99"
"99"
"1235"
"1235"
"1235"
"17-3021"
"17-2020"
"17-2011"

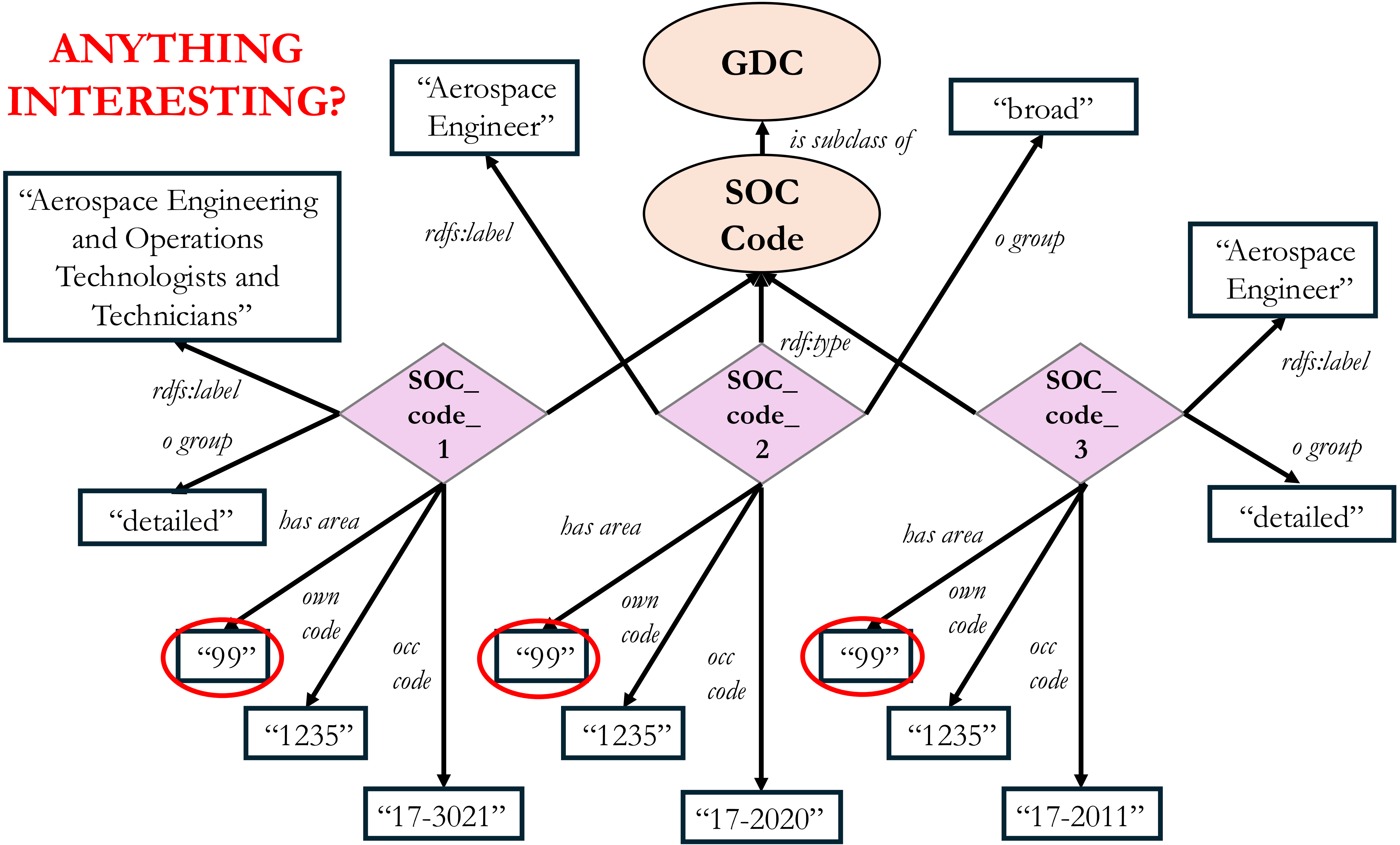


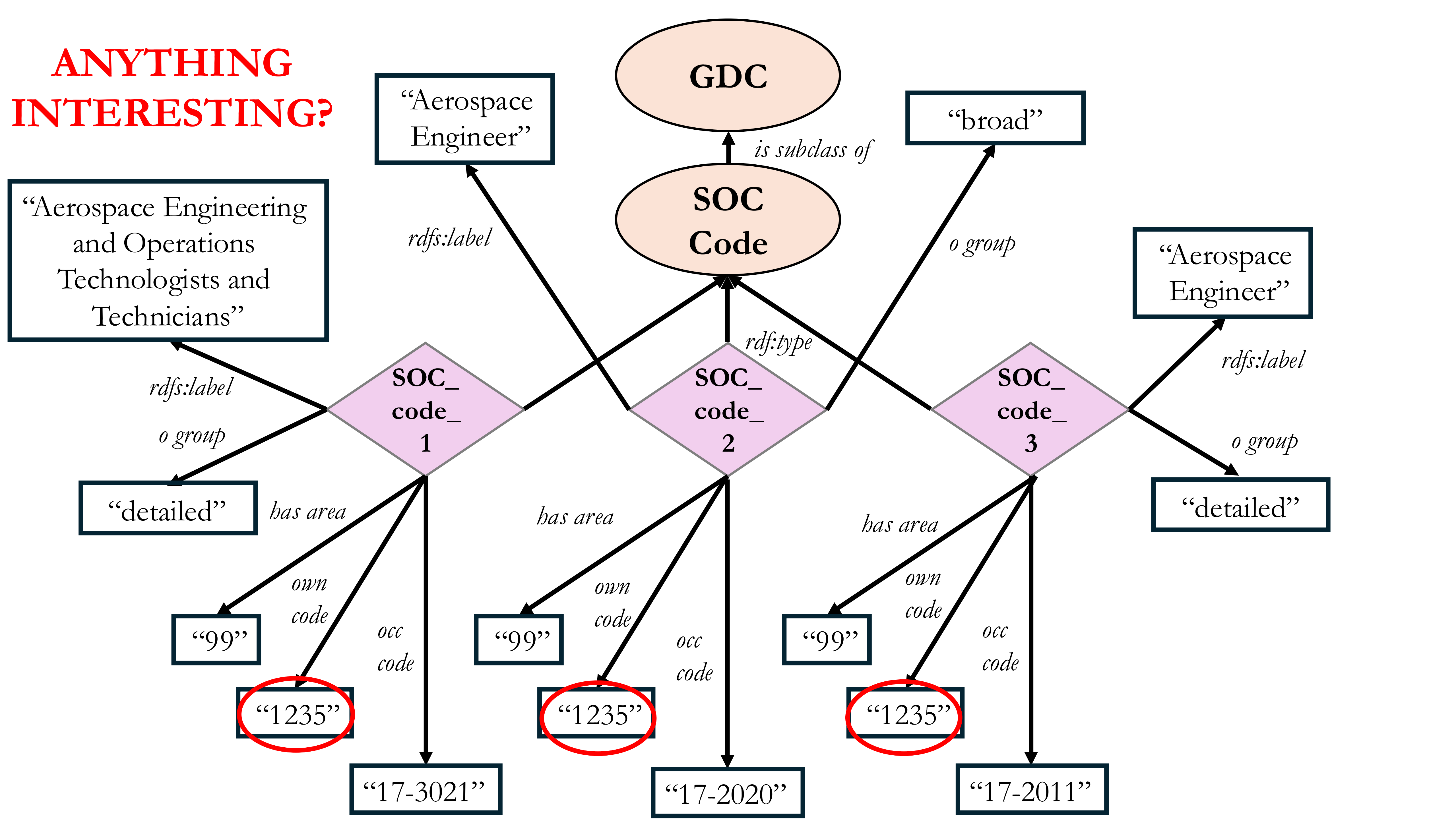
MORE ANNOTATIONS
GDC
is subclass of
SOC Code
rdf:type
"Aerospace Engineer"
"broad"
"Aerospace Engineering and Operations Technologists and Technicians"
"Aerospace Engineer"
rdfs:label
o group
SOC_ code_ 1
SOC_ code_ 2
SOC_ code_ 3
"detailed"
"detailed"
has area
own code
occ code
"99"
"1235"
"17-3021"
"99"
"1235"
"17-2020"
"99"
"1235"
"17-2011"

ANYTHING INTERESTING?

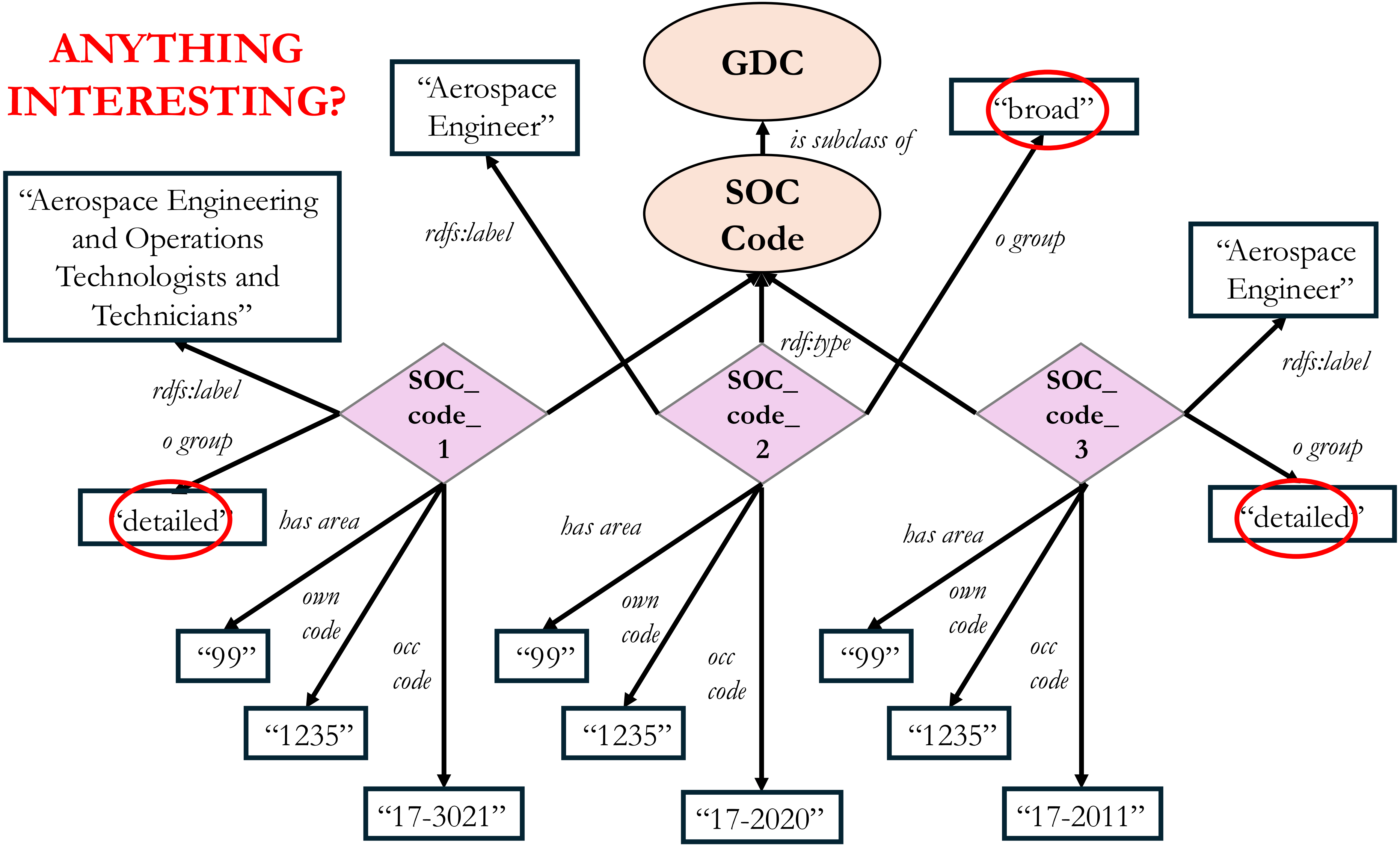


ANYTHING INTERESTING?

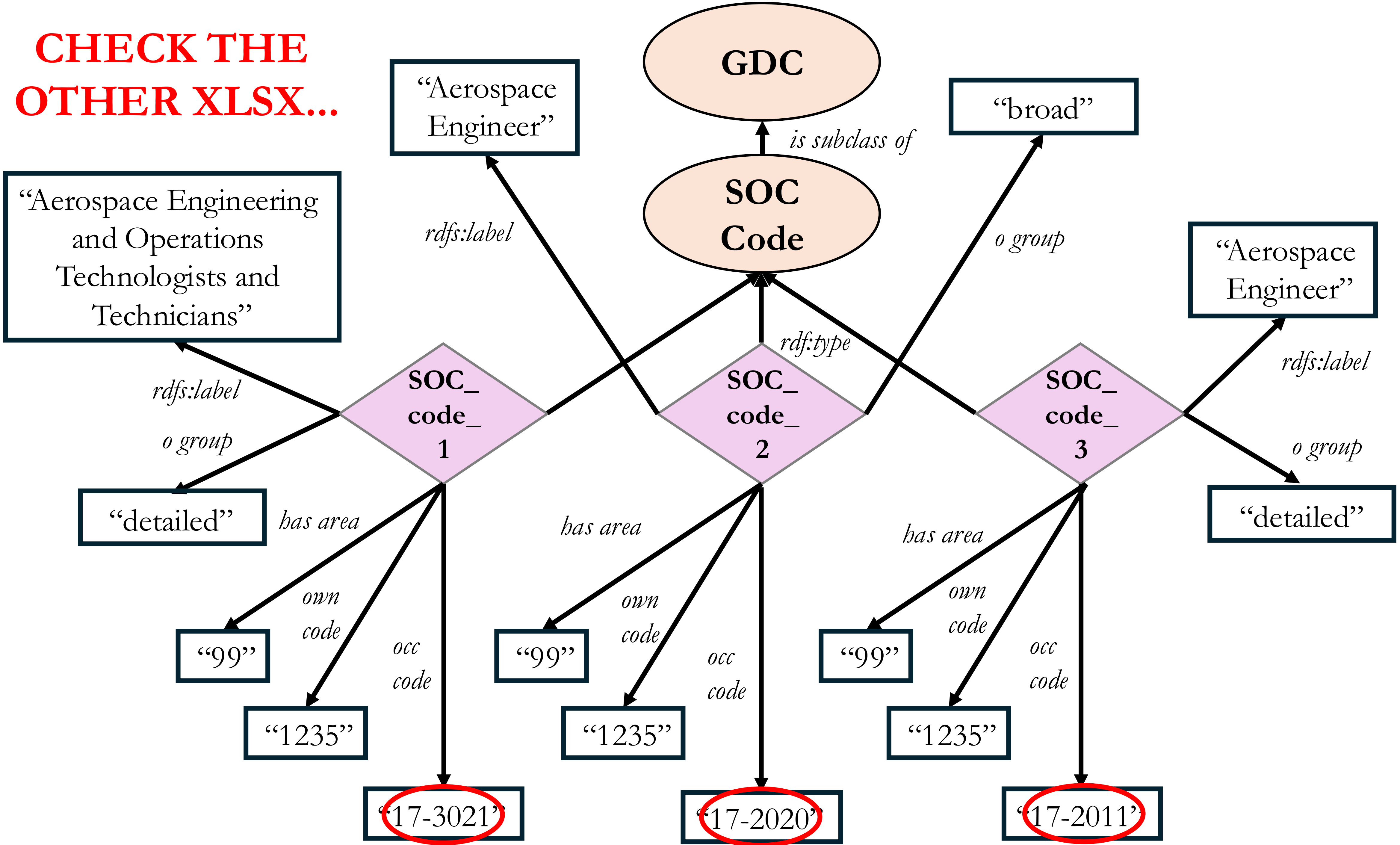


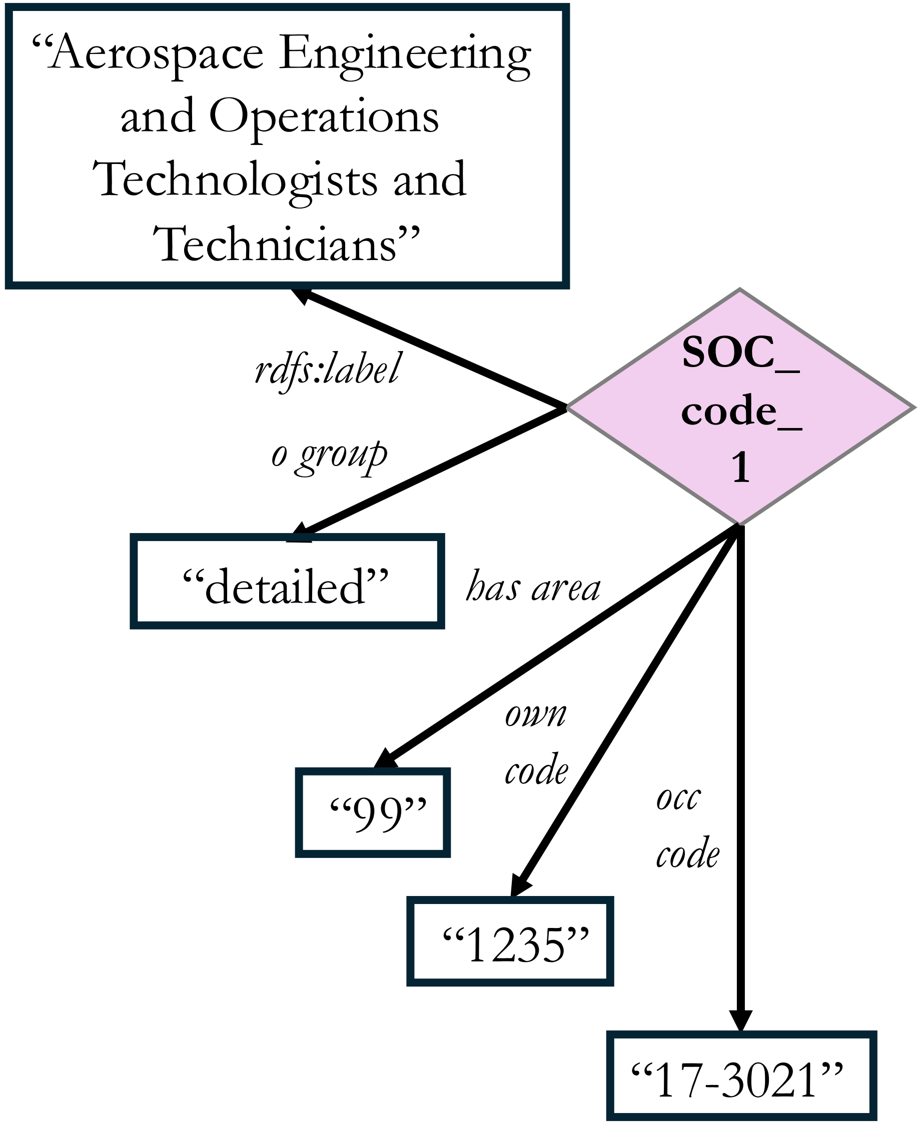

ANYTHING INTERESTING?
GDC
is subclass of
SOC Code
"Aerospace Engineer"
"broad"
"Aerospace Engineering and Operations Technologists and Technicians"
rdfs:label
o group
rdf:type
"Aerospace Engineer"
rdfs:label
SOC_ code_ 1
SOC_ code_ 2
SOC_ code_ 3
rdfs:label
o group
"detailed"
has area
has area
has area
o group
"detailed"
own code
own code
own code
occ code
occ code
occ code
"99"
"99"
"99"
"1235"
"1235"
"1235"
"17-3021"
"17-2020"
"17-2011"

ANYTHING INTERESTING?

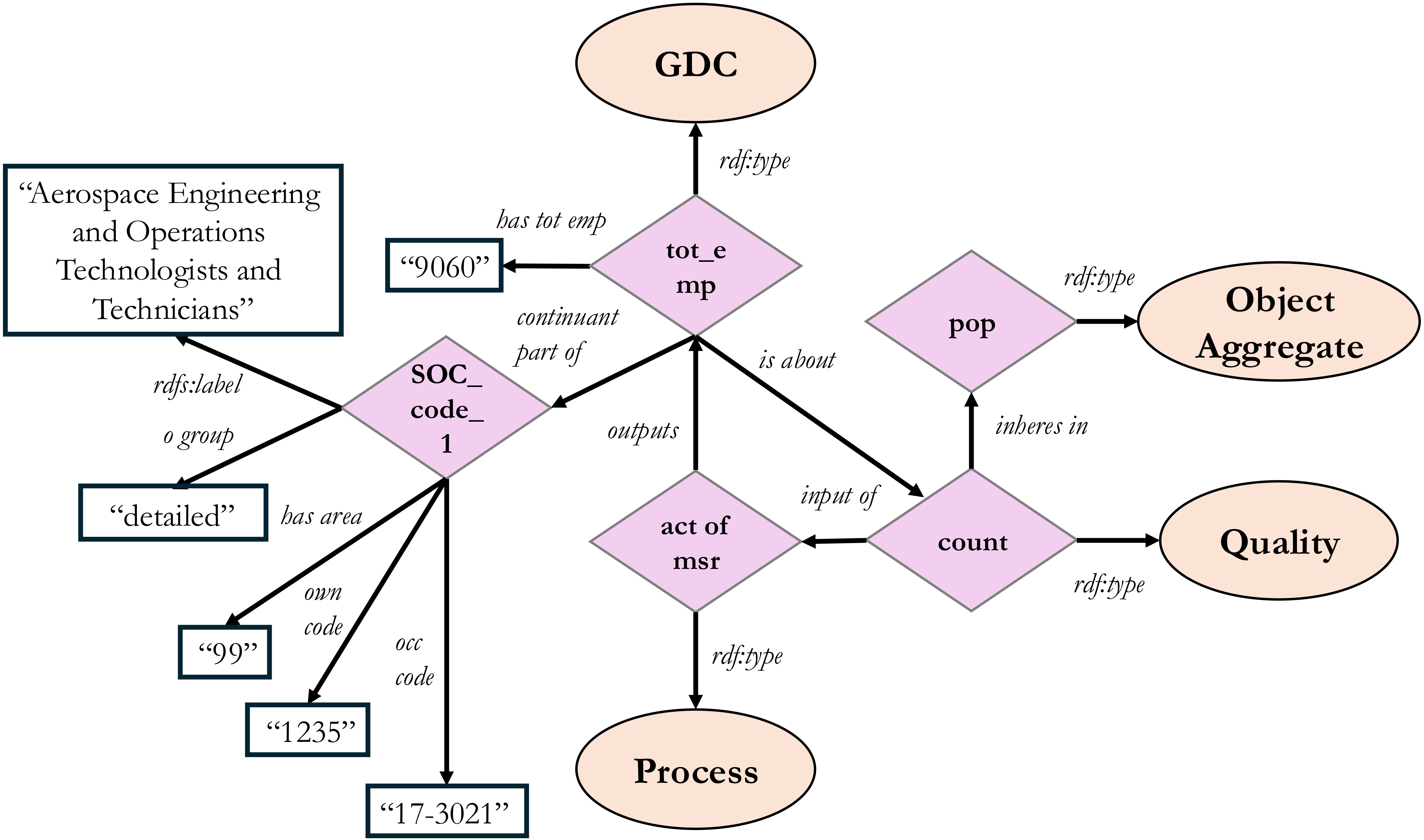


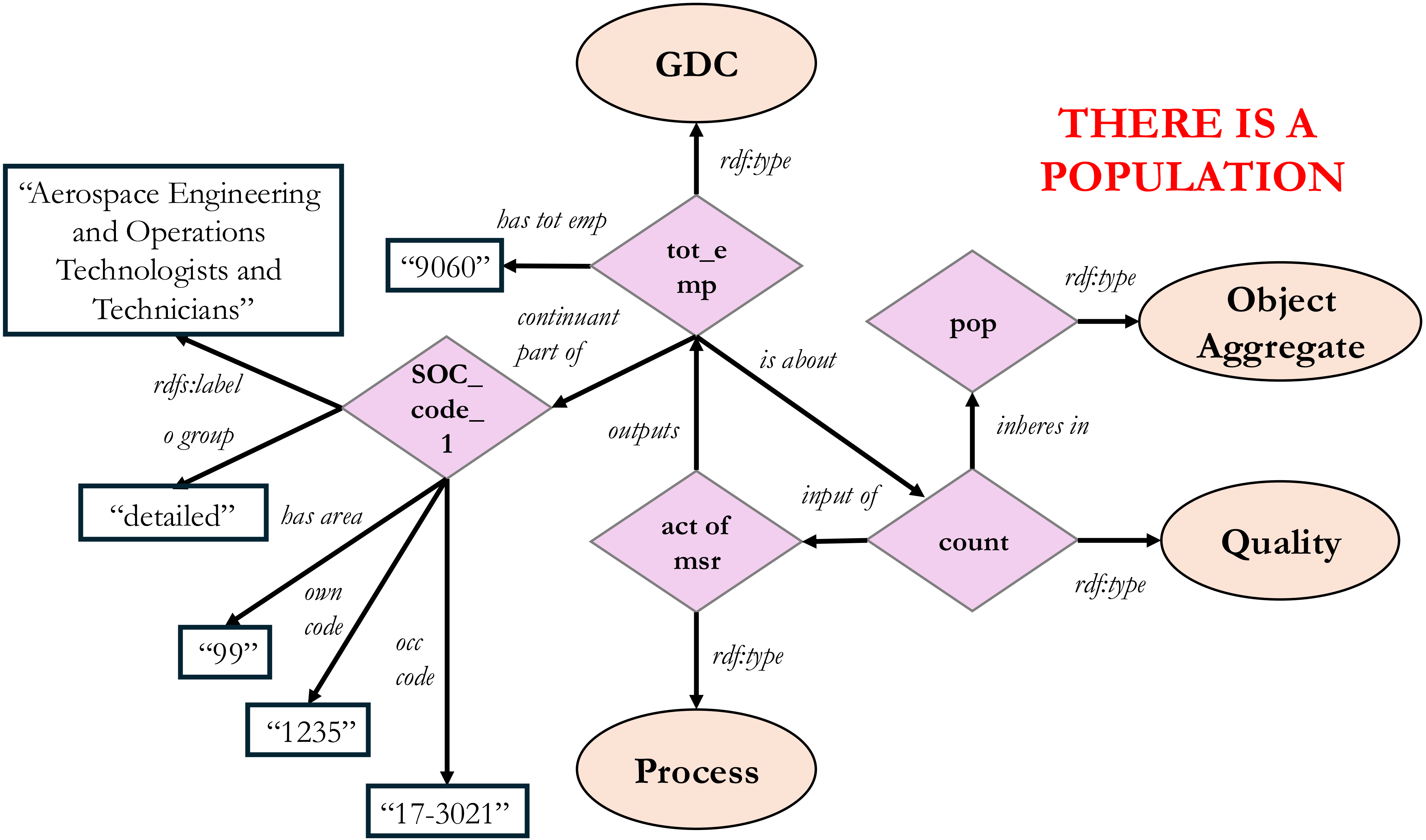
CHECK THE OTHER XLSX…





“Aerospace Engineering and Operations Technologists and Technicians”
rdfs:label
o group
SOC_ code_ 1
“detailed”
has area
own code
occ code
“99”
“1235”
“17-3021”


GDC
rdf:type
has tot emp
tot_e
mp
"9060"
"Aerospace Engineering and Operations Technologists and Technicians"
continuant part of
rdf:type
pop
Object Aggregate
is about
SOC_
code_
1
rdfs:label
o group
outputs
inheres in
"detailed"
has area
input of
act of
msr
count
Quality
own code
rdf:type
"99"
occ code
rdf:type
"1235"
Process
"17-3021"


THERE IS A POPULATION
GDC
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tot_emp
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"Aerospace Engineering and Operations Technologists and Technicians"
continuant part of
is about
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SOC_code_1
o group
"detailed"
has area
own code
"99"
"1235"
occ code
"17-3021"
outputs
act of msr
input of
count
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Object Aggregate
inheres in
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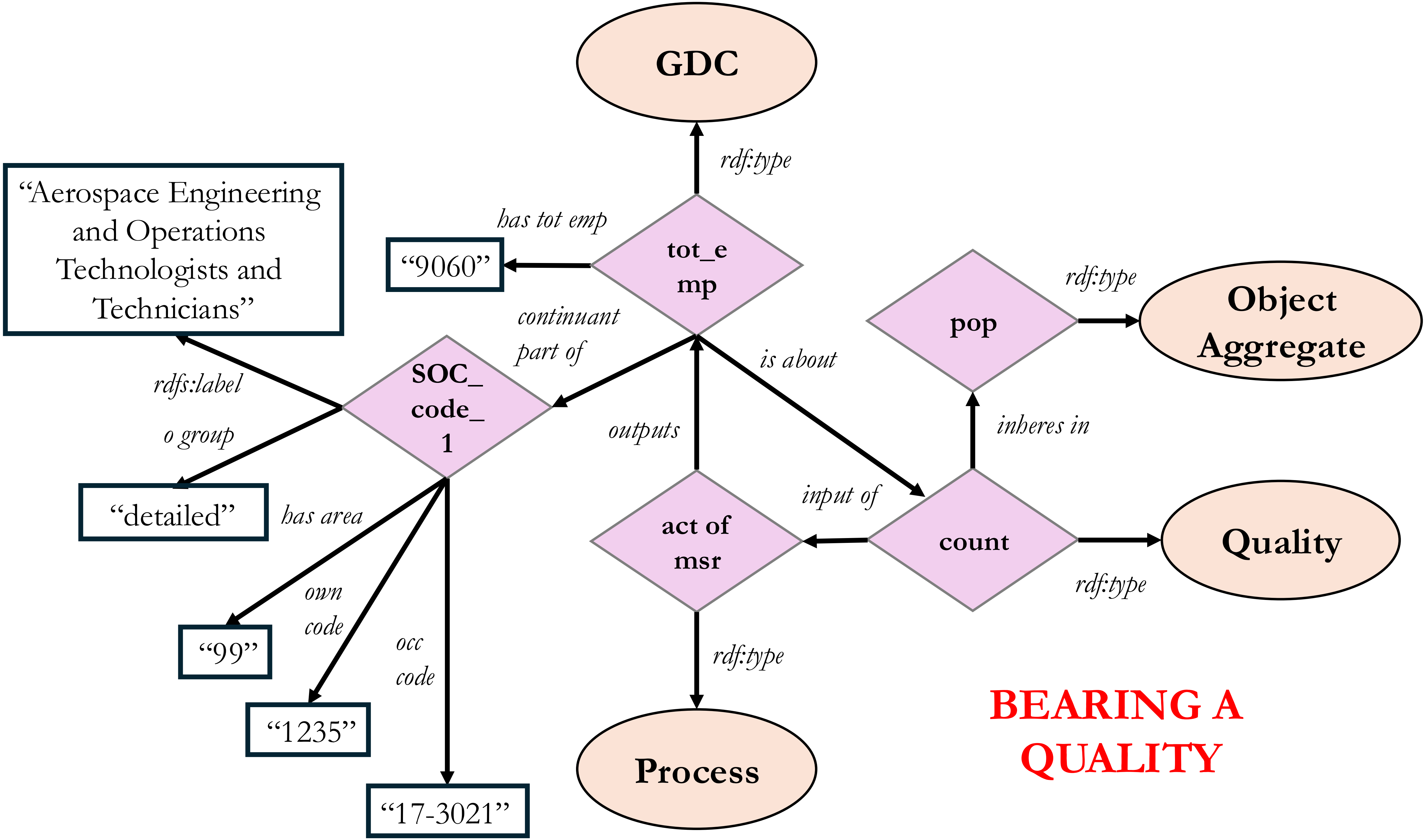
Process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has tot emp
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tot_e
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is about
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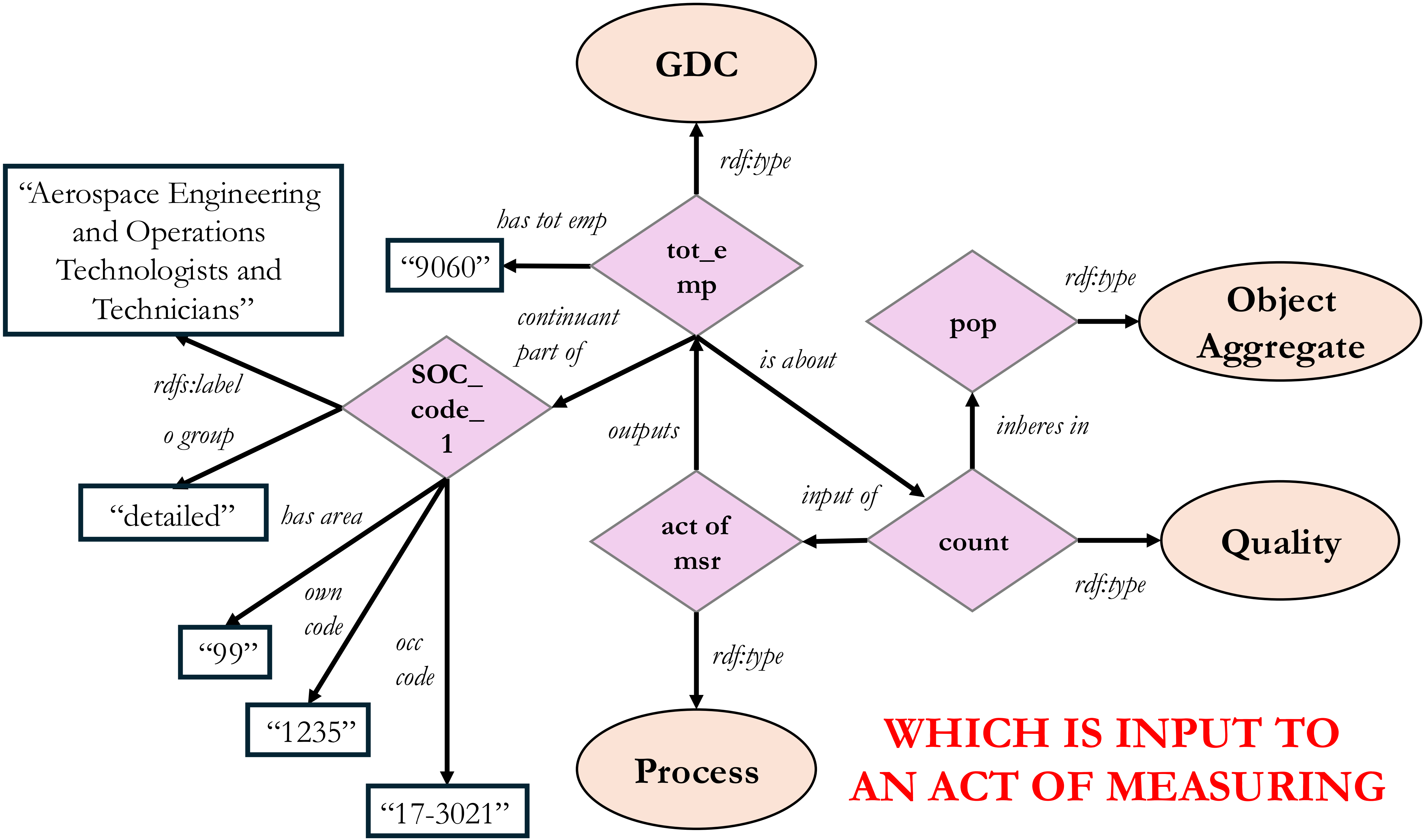
rdf:type
Object
Aggregate
rdfs:label
SOC_
code_
1
outputs
inheres in
o group
"detailed"
has area
input of
act of
msr
count
Quality
rdf:type
own
code
"99"
occ
code
rdf:type
"1235"
BEARING A
QUALITY
Process
"17-3021"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SOC_
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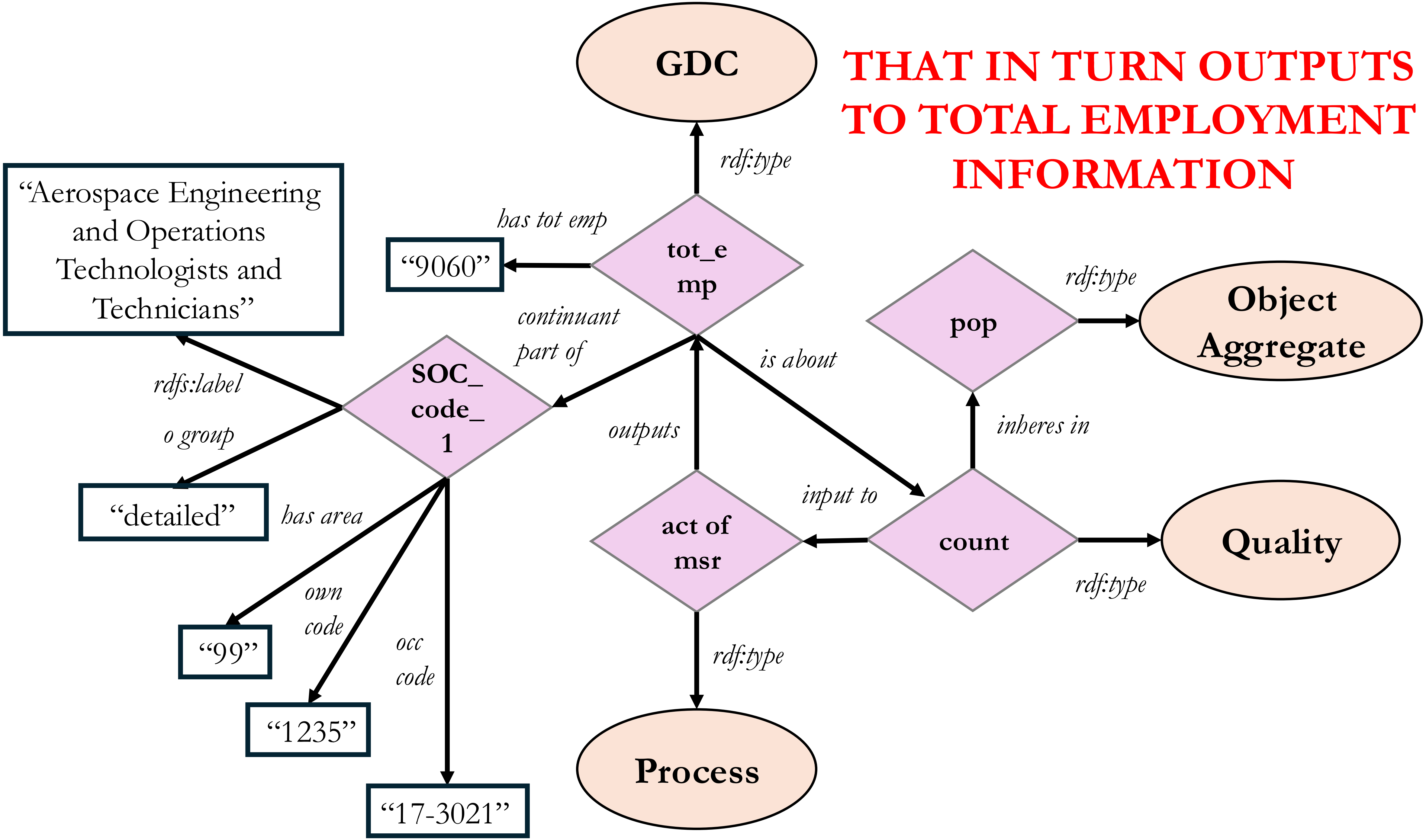
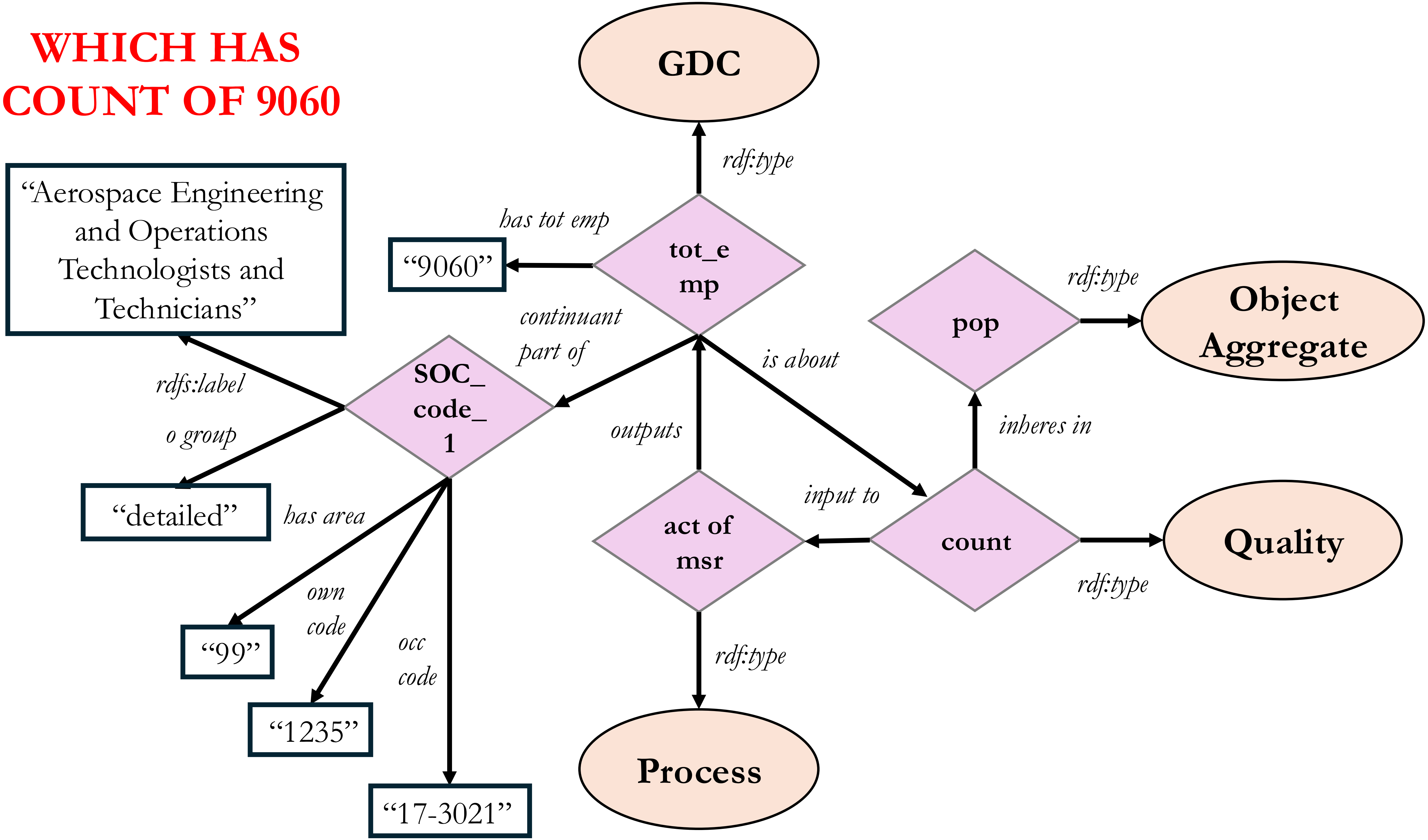
o group
"detailed"
has area
own code
occ code
"99"
"1235"
"17-3021"
outputs
is about
pop
rdf:type
Object Aggregate
inheres in
input of
act of
msr
count
rdf:type
Quality
rdf:type
Process
WHICH IS INPUT TO AN ACT OF MEASURING

THAT IN TURN OUTPUTS TO TOTAL EMPLOYMENT INFORMATION



WHICH HAS COUNT OF 9060



AND IS CONTINUANT PART OF THE SOC CODE